The Mechanics of Delta Learning: Target Design for Generalizable Scientific Machine Learning

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Abstract

Delta learning—training machine learning models on residual errors relative to baseline physical approximations—is often used across scientific domains to predict complex physical systems. However, baseline selection remains largely empirical, governed by the assumption that increasingly accurate baselines with smaller residual scales inherently improve downstream performance. Here, we show that residual scale alone is an insufficient heuristic for learnability and introduce a two-dimensional target-design framework that evaluates baseline transformations across both scale reduction and model-relative smoothness. Investigating molecular graph neural network (GNN) architectures on total energy targets reveals that more accurate local descriptor baselines can leave smaller residual targets that are disproportionately rough within architecture-informed proxy spaces and harder to learn relative to their scale. In contrast, an xTB baseline reduces both scale and normalized roughness while improving in-domain and out-of-domain prediction. A DFTB control improves prediction despite a small increase in normalized roughness, showing that smoothness is a diagnostic rather than a necessary condition for lower absolute error. We introduce scale-normalized graph Dirichlet roughness (D_{IQR}) as a candidate pre-training diagnostic for residual learnability. Its association with normalized test error contains a substantial shared-normalization contribution, with permutation controls supporting additional molecular-neighbourhood information. These results motivate baseline complementarity as a target-design principle, positioning target space formulation alongside model architecture as an important axis for scientific machine learning.

1 Introduction

Supervised machine learning maps an input representation to a target property. The generalizability of this mapping fundamentally depends on its smoothness; i.e., the degree to which small, continuous changes in the feature space produce small changes in the target property. Standard model development focuses primarily on the representation side of this problem, refining architectures and inductive biases to build feature spaces where the target varies smoothly [1, 2, 3, 4, 5]. The prediction target, by contrast, is usually treated as fixed.

However, heterogeneous targets, such as molecular total energies, combine compositional, local, many-body, and long-range electronic contributions that span different interaction orders and length scales. A fixed graph neural network (GNN) architecture may represent these contributions with different accuracy. Poorly aligned components may therefore increase target roughness within the model's feature space and limit downstream generalizability.

Herein, we propose target design as an approach to improve generalization by considering target scale and smoothness within an architecture-informed representation space. We execute this transformation through delta learning, testing baselines that may complement the model's inductive bias [6]. Crucially, this transformation alters both target scale and functional form. At fixed normalized error, shrinking target scale scales down the absolute error; changes in functional form can also change scale-normalized learnability.

Contradictory results in the delta-learning literature motivate this view. Delta learning is commonly motivated by the expectation that a baseline with greater accuracy, stronger correlation with the parent target, or better alignment with its dominant variation will remove more target variation, leaving a smaller residual that is assumed to be easier to learn [7, 8, 9, 10]. Under this view, better baselines should produce smaller residuals and better downstream models. Yet, other studies report negative results or show that less accurate baselines outperform more accurate ones [11, 12]. Baseline accuracy and residual scale are therefore insufficient; the residual's learnability relative to the model's inductive bias must also be considered.

To evaluate model-relative learnability post-training, prediction error must be separated from target scale. We perform this analysis using normalized MAE (n-MAE), defined as the prediction MAE divided by the target interquartile range (IQR). An increase in n-MAE indicates that the residual is harder for the model to learn relative to its scale. The most favorable transformation reduces target scale while maintaining or improving normalized learnability.

Because n-MAE is available only post-training, target design benefits from a diagnostic of representation-space smoothness that does not require downstream training. We introduce IQR-normalized graph Dirichlet roughness, D_{IQR}, which directly quantifies target variation across a fixed feature space informed by the downstream architecture [13]. Low Dirichlet roughness indicates that molecules mapped close together in feature space have similar target values relative to scale, motivating its evaluation as a candidate pre-training screening diagnostic.

We evaluate this framework on molecular total energies using GemNet-T across a baseline hierarchy spanning varied physical priors, accuracies, and degrees of feature-space alignment [14]. We also perform parallel analyses using SchNet, contrasting its pair-distance representation with GemNet-T's directional and angular message-passing scheme [4]. In-domain performance is benchmarked on organometallic complexes from tmQM, and out-of-distribution transferability is assessed on organic molecules from QM9 [15, 16].

Across both architectures, replacing linear descriptor baselines with more accurate MLP fits leaves smaller but markedly rougher residuals. Normalized roughness is associated with post-training n-MAE across the primary target panel, although shared normalization contributes substantially to this relationship. Conversely, the xTB baseline reduces target scale and normalized roughness

while improving in-domain accuracy and out-of-distribution transfer. DFTB supplies a supporting control that improves prediction despite slightly increased normalized roughness. Delta learning can therefore be framed as a two-dimensional target-design problem involving residual scale and representation-space smoothness, motivating baseline assessment beyond baseline accuracy alone.

2 Results

2.1 Target shrinkage does not guarantee learnability

To examine whether target shrinkage and model-relative learnability are equivalent, we first test a standard target-preprocessing protocol used in machine-learning interatomic potentials (MLIPs) [17, 4, 18]. These reference corrections are widely used to improve optimization and training stability. First, fixed element-wise atomic references are subtracted from the total DFT energy to yield the reference-corrected energy, E_{ref}. Next, a linear Bag-of-Elements (BoE) model fitted to elemental counts is subtracted, defining the residual target:

$$E_{\mathrm{ref,BoE}} = E_{\mathrm{ref}} - E_{\mathrm{BoE}}. \tag{1}$$

Because the subtracted references are added back at inference, this procedure constitutes stacked delta learning. The MAE of the residual prediction is exactly equal to that of the complete stacked prediction (SI Section 1).

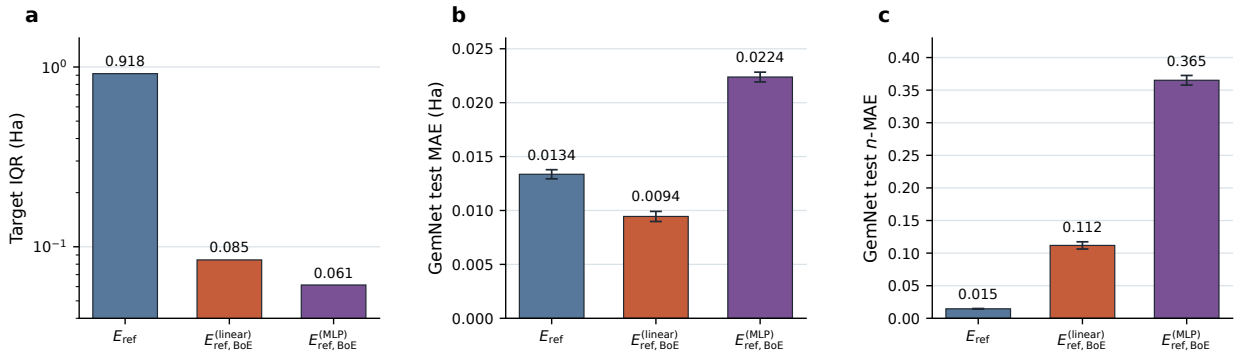


Figure 1: **Target scale and GemNet accuracy after BoE correction. a,** Test-set IQR for E_{ref} and its linear- and MLP-BoE residuals. **b,** Mean GemNet test MAE. **c,** Normalized MAE, $n\text{-MAE}_{\mathrm{test}} = \mathrm{MAE}_{\mathrm{test}} \,/\, \mathrm{IQR}_{\mathrm{test}}(y)$. Error bars show sample standard deviations across three seeds.

In addition to its conventional role in improving optimization and training stability, linear BoE correction reduces the mean tmQM test MAE by a factor of 1.41 (Fig. 1b). BoE correction is therefore a successful form of delta learning. However, this improvement could arise either from target shrinkage or from a transformation that makes the residual better aligned with GemNet's inductive bias and, consequently, easier to learn relative to its scale.

To distinguish these effects, we use the normalized MAE, $n\text{-MAE}_{\mathrm{test}} = \mathrm{MAE}_{\mathrm{test}} \,/\, \mathrm{IQR}_{\mathrm{test}}(y)$, where the interquartile range (IQR) measures the central scale of the test target. The n-MAE provides an empirical, model-dependent measure of learning difficulty after accounting for target scale. Under a pure rescaling, MAE and IQR change by the same factor, leaving n-MAE unchanged. An algebraic rescaling control confirms this invariance (SI Note 2). An increase in n-MAE after a target transformation therefore indicates that the transformed target is harder for the same model to learn relative to its own scale, whereas a decrease indicates improved model-relative learnability.

For linear BoE correction, the n-MAE of $E_{\mathrm{ref,BoE}}$ is 7.7-fold larger than that of E_{ref} (Fig. 1c), while the target IQR decreases by a factor of 10.9 (Fig. 1a). The reduction in absolute MAE therefore does not result from improved model-relative learnability. Instead, the substantial target shrinkage outweighs the increase in normalized learning difficulty.

The linear BoE result demonstrated that substantial target shrinkage can improve absolute performance despite reducing normalized learnability. To test whether further target shrinkage continues to improve downstream performance, we replace linear BoE with a nonlinear MLP fitted to the same elemental-count inputs. This control increases baseline flexibility while holding its input information fixed. The MLP residual is 1.38-fold smaller by IQR, but its n-MAE is 3.27-fold larger, resulting in a 2.37-fold increase in absolute MAE (Fig. 1). SchNet shows the same trend (SI Section 1). This result rejects the assumption that progressively removing more target variation necessarily produces progressively better residual learning.

These results establish the central problem addressed in this work: baseline accuracy and its accompanying residual scale reduction alone cannot predetermine the success of a target transformation. While the normalized error assesses scale-independent learnability of the target, it remains a post-hoc metric only available after training. A practical target-design strategy therefore requires a pre-training measure that estimates how a baseline transformation changes the normalized learnability of its residual.

2.2 Pre-training roughness quantifies model-relative learnability

To evaluate whether a baseline residual will be learnable downstream, we measure its smoothness relative to structural information accessible to the target neural network. However, because the trained internal representations of message-passing networks such as SchNet and GemNet are learned during optimization, they are unavailable *a priori*. We address this by constructing fixed, explicit geometric feature spaces, termed **representation proxies**, that approximate the geometric information available to each architecture:

- **Pair-Radial Expansion (PRE):** a two-body feature space that expands element-resolved pairwise interatomic distances, r_{ij}, using Gaussian radial basis functions up to a local spatial cutoff. By capturing distance distributions without explicit angular information, PRE serves as a representation proxy for pairwise message-passing architectures such as SchNet.
- **PRE with angular terms (PRE+A):** an extended three-body feature space that augments PRE with central-element-resolved triplet angle distributions, θ_{ijk}. By capturing angular geometry in addition to pairwise distances, PRE+A serves as a proxy for directional, triplet-based architectures such as GemNet.

Detailed descriptor formulations, radial basis dimensions, and hyperparameter choices are provided in Methods (Sections 5.2 and 5.4) and SI Methods.

By mapping molecules into these proxy feature spaces, we can evaluate target variation before model optimization. We construct a k-nearest-neighbour graph connecting test molecules with similar standardized descriptor vectors. Edge weights, w_{ij}, encode feature-space similarity through a locally scaled Gaussian kernel of cosine distance:

$$w_{ij} = \exp\left(-\frac{d_{ij}^2}{\sigma_i \sigma_j}\right), \tag{2}$$

Here, d_{ij} is the cosine distance between standardized descriptor vectors for molecules i and j, and σ_i is a local distance scale, defined as the distance from molecule i to its k-th nearest neighbour.

Using these edge weights, we quantify target variation through the IQR-normalized graph Dirichlet roughness, D_{IQR} [13]:

$$D_{\mathrm{IQR}}(y) = \frac{1}{\sum_{\{i,j\}\in\mathcal{E}} w_{ij}} \sum_{\{i,j\}\in\mathcal{E}} w_{ij} \frac{(y_i - y_j)^2}{\mathrm{IQR}_{\mathrm{test}}(y)^2}. \tag{3}$$

Here, $\mathcal{E}$ is the set of unique undirected graph edges. For targets with nonzero IQR, this formulation normalizes local squared target differences by the squared central target spread, making D_{IQR} invariant to nonzero uniform target rescaling. A lower value indicates that molecules with similar proxy descriptors have similar target labels relative to this central spread. This provides an architecture-informed, pre-training diagnostic for assessing residual learnability without first training a downstream GNN; its predictive value is tested in the following comparisons.

Applying this normalized roughness metric to the composition-corrected energy targets, we find that D_{IQR} increases from 0.548 for E_{ref} to 1.127 for linear-BoE and 1.658 for MLP-BoE. This ordering matches the post-training n-MAE ordering observed in Figure 1. The MLP-BoE residual is similarly rougher than linear-BoE in the SchNet-aligned PRE space (1.709 versus 1.152; SI Note 6).

In the following sections, we further test whether D_{IQR} can serve as a reliable qualitative ordering proxy for target roughness and whether that roughness describes the intrinsic learnability assessed post-training using n-MAE.

2.3 Residual roughness guides target design beyond baseline accuracy

To test whether D_{IQR} reliably predicts downstream learnability, we systematically evaluated how different baseline physical approximations alter residual scale, roughness, and deep model performance. Our goal is to construct simple, computationally inexpensive baseline models that predict the atom- and composition-referenced DFT energy, $E^{\mathrm{DFT}}_{\mathrm{ref,BoE}}$. Subtraction of these predictions, $\widehat{E}_b$, yields the transformed residual target, Δ^b, on which the downstream GNN models are trained.

We compare three descriptor-based baselines, BoB, PRE, and PRE+A, that capture progressively richer structural information. Alongside these, we evaluate xTB as our primary semiempirical comparison and DFTB as a supporting control. Table 1 summarizes the five approaches:

1. **Bag-of-Bonds (BoB):** a simple global descriptor tracking RDKit-derived typed-bond counts to capture basic compositional and functional-group variation, rather than the Coulomb-matrix entries used in the conventional BoB descriptor.

2. **Pair-Radial Expansion (PRE):** repurposing the pairwise geometric representation introduced in Section 2.2 as an explicit input descriptor to capture local two-body geometry.

3. **PRE with angular terms (PRE+A):** repurposing the three-body geometric representation from Section 2.2 as an input descriptor to capture local pairwise and angular geometry.

4. **Extended tight-binding (xTB):** a semiempirical quantum-mechanical baseline providing molecule-wide electronic-structure information beyond the local geometric cutoffs of the descriptors.

5. **Density functional tight-binding (DFTB†):** a second semiempirical quantum-mechanical baseline used to test whether the trends observed with xTB extend to another electronic-structure method.

Table 1: **Baseline models used for target transformation.**

Baseline model	Input features	Fitting approach	Physical/representation target
BoB	Typed-bond counts (RDKit)	Linear / MLP	Compositional and functional-group variation
PRE	Pairwise distances (r_{ij})	Linear / MLP	Local two-body geometry (SchNet proxy)
PRE+A	Distances and angles (r_{ij}, θ_{ijk})	Linear / MLP	Local three-body geometry (GemNet proxy)
xTB	Atomic identities and molecular coordinates	Semiempirical QM	Non-local electronic and higher-order interactions
DFTB†	Atomic identities and molecular coordinates	Semiempirical QM	Molecule-wide electronic structure; supporting comparison with xTB

For each descriptor (BoB, PRE, and PRE+A), we fit both a linear regression model and a nonlinear multilayer perceptron (MLP); the MLPs share a unified hidden architecture (Methods). This dual-fitting strategy enables two controlled evaluations: varying the input descriptor under a fixed MLP architecture assesses how baseline information content alters residual structure, while comparing linear and MLP fits within the same descriptor family isolates the effect of baseline fitting capacity. For each supervised baseline b, the target residual is defined as:

$$\Delta^b = E_{\mathrm{ref,BoE}}^{\mathrm{DFT}} - \widehat{E}_b, \tag{4}$$

In parallel, we evaluate xTB to provide a physics-based baseline that accounts for non-local electronic interactions [19]. Its residual, Δ^{xTB}, is computed after applying the same atom- and composition-referencing protocol:

$$\Delta^{\mathrm{xTB}} = E_{\mathrm{ref,BoE}}^{\mathrm{DFT}} - E_{\mathrm{ref,BoE}}^{\mathrm{xTB}}. \tag{5}$$

The DFTB residual is constructed in the same way, using the atom- and composition-referenced DFTB energy [20, 21]. DFTB calculations failed for 362 neutral tmQM molecules, so this control uses the successful-calculation subset and one GNN seed per architecture. DFTB† is shown in the main figures for comparison but excluded from pooled correlations and regression fits. Matched-subset comparisons and calculation details are provided in SI Note 8.

Comparing MLP baselines across descriptor families reveals a clear trade-off between target accuracy and learnability. Progressing from BoB to PRE and PRE+A progressively increases baseline accuracy and reduces residual magnitude (Fig. 2a,c). Judged strictly by baseline accuracy or residual scale, PRE+A would appear to offer the optimal Δ-learning target. However, our roughness analysis yields a different conclusion. Although these baselines reduce target IQR by factors of 3.3, 4.2, and 4.7, respectively, they increase representation-space target roughness (D_{IQR}) by factors of 3.3, 9.5, and 7.6 within GemNet-aligned PRE+A space (Fig. 2b). Consequently, these smaller residuals remain rougher, and none improves GemNet test MAE over direct training on $E_{\mathrm{ref,BoE}}$.

This degradation in learnability persists even when baseline input features remain fixed. Comparing linear and MLP fits within each descriptor family shows that increasing baseline expressivity consistently yields smaller yet rougher residuals, degrading GemNet test performance (Fig. 2). For instance, transitioning from a linear to an MLP fit on BoB features reduces residual IQR by an additional factor of 1.6, yet GemNet test MAE increases 1.7-fold. Thus, stripping away

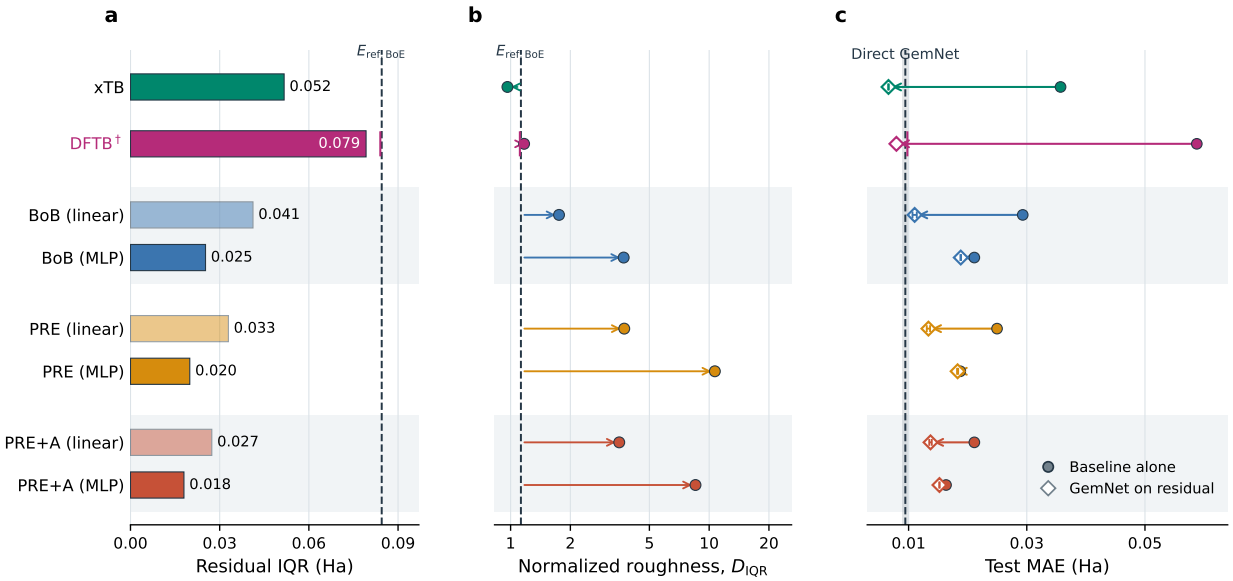


Figure 2: **Baseline accuracy, residual roughness, and GemNet test performance. a,** Test residual IQR. **b,** Normalized roughness on the test PRE+A graph (log scale). **c,** Baseline-only test MAE (filled circles) and GemNet residual test MAE (open diamonds). Dashed lines mark the DFT-BoE parent or direct-GemNet reference; arrows show the corresponding changes. Main GNN results are three-seed means; error bars and shading show sample SD. Magenta DFTB† is a single-seed, valid-subset control, excluded from pooled analyses. Short magenta ticks mark its matched-subset parent references (SI Note 8).

additional target variance without considering representation alignment can actively impede model generalization.

The poorer test performance on these rough residuals persists after normalization by test target IQR. For GemNet, the MLP-PRE residual has a normalized test error 7.2-fold higher than the xTB residual. SchNet exhibits a similar discrepancy, with a 4.9-fold difference (SI Note 6). Thus, even after accounting for target scale, the smaller MLP-PRE residual remains harder to predict on unseen structures.

In contrast, xTB leaves a larger residual than the BoB, PRE, and PRE+A baselines, yet GNNs trained on this residual achieve lower test errors. Among the primary baselines in Figure 2, xTB is the only one that lowers both normalized roughness and GemNet test MAE: D_{IQR} falls from 1.127 to 0.964, while mean test MAE falls from 0.00945 to 0.00660 Ha. xTB therefore provides our primary evidence that smaller residuals are not necessarily easier for a GNN to learn.

DFTB provides a contrasting control on its valid subset: its residual is also relatively large, and has slightly higher roughness than the matched DFT parent in both PRE and PRE+A spaces. In PRE+A space, D_{IQR} rises from 1.113 to 1.169, while the matched seed-101 GemNet test MAE falls from 0.00979 to 0.00794 Ha (SI Note 8). DFTB remains a separate semiempirical control, not part of the pooled analysis. These results do not suggest that larger residuals are inherently better. Rather, they show that an effective baseline must leave a residual that the downstream model can learn, not simply one with a smaller magnitude.

2.4 Model-relative target roughness predicts scale-normalized learnability

Can target roughness measured prior to training identify which targets a GNN will learn more accurately? We address this by comparing representation-space roughness (D_{IQR}) with scale-normalized test error (n-MAE $= \mathrm{MAE}_{\mathrm{test}} \,/\, \mathrm{IQR}_{\mathrm{test}}(y)$). Graph-based smoothness provides a geometric framework for relating labels to neighbouring representations.[13] Normalized error assesses how accurately a model predicts each target relative to that target's intrinsic spread, enabling comparisons across energy targets with widely different scales.

We evaluated all 13 primary targets: atom-referenced DFT energy, linear and MLP Bag-of-Elements (BoE) residuals, referenced xTB energies and their DFT residuals, and linear and MLP residuals derived from BoB, PRE, and PRE+A representations. Because the proxy feature graph remains fixed across targets for a given architecture, differences in D_{IQR} reflect how target values vary relative to their spread between neighbouring molecules, rather than changes in graph connectivity.

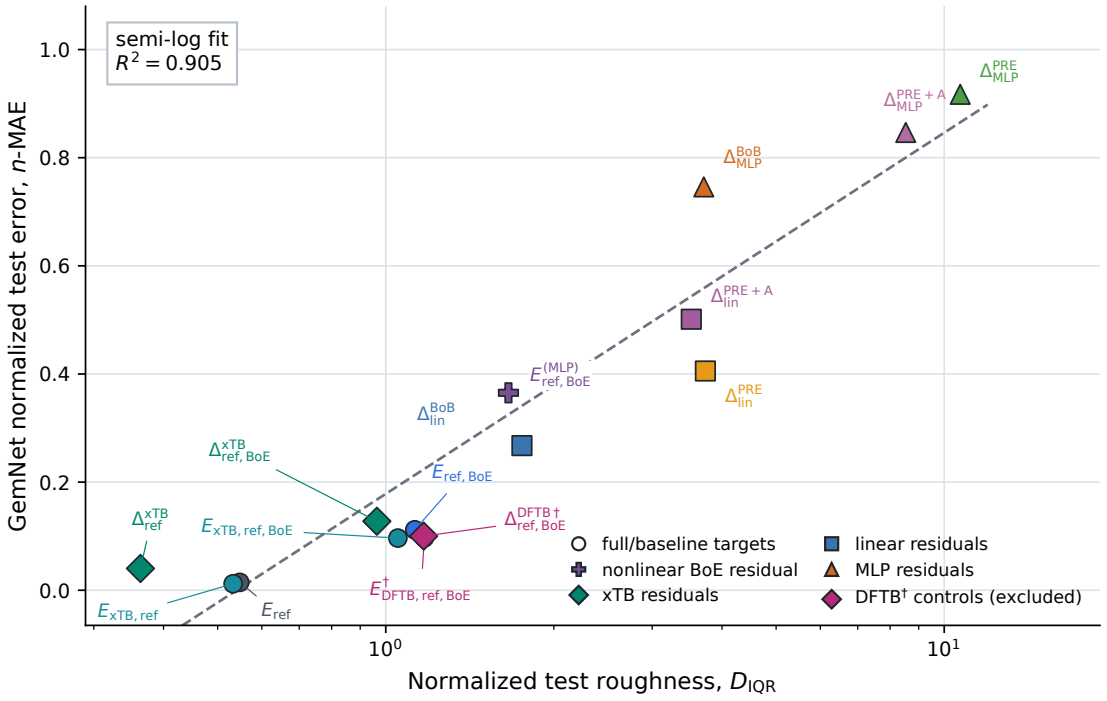


Figure 3: **Model-relative roughness and normalized GemNet test error.** Mean GemNet test n-MAE, defined as $\mathrm{MAE}_{\mathrm{test}} \,/\, \mathrm{IQR}_{\mathrm{test}}(y)$, is plotted against test D_{IQR} for 13 primary targets. Roughness is measured on the standardized test PRE+A graph, and the horizontal axis is logarithmic. Marker shapes identify the target classes shown in the legend. Points show means across three GNN seeds. The dashed line is the semi-log regression over the 13 primary target means ($R^2 = 0.905$). Magenta DFTB† points are single-seed, valid-subset controls, excluded from the regression and correlations.

Across these primary targets, GemNet's normalized test error increases approximately linearly with $\log D_{\mathrm{IQR}}$, yielding $R^2 = 0.905$ in PRE+A space (Fig. 3). SchNet exhibits a corresponding association in PRE space ($R^2 = 0.946$; SI Note 6). Practically, targets that vary more sharply between structurally similar molecules tend to be harder for downstream GNNs to learn relative to their spread. Leave-one-target-out sensitivity checks show that this trend is not driven by any single target, maintaining $R^2 \in [0.872, 0.927]$ for GemNet and $R^2 \in [0.927, 0.965]$ for SchNet (SI

Note 7). These ranges describe refitted associations, not cross-validated prediction scores.

Because D_{IQR} and n-MAE share target IQR as a scaling factor, their correlation may partly reflect shared-denominator normalization.[22] We conducted sensitivity controls to assess the contribution of molecular-neighbourhood alignment beyond this coupling. Removing IQR normalization from both axes reduces R^2 to 0.019 for GemNet and 0.067 for SchNet, showing that the strong normalized association does not extend to the relationship between log raw roughness and absolute test error (SI Note 7).

To assess the contribution of label alignment with molecular graph geometry, we performed 5,000 joint-label permutations (SI Note 7). Shuffling target values across molecules preserves target distributions and IQRs while severing their connection to chemical structure; graphs and GNN test errors remain fixed. Although target distributions and normalization sustain a high shuffled correlation (median $R^2 = 0.862$ for GemNet and 0.855 for SchNet), the observed positive correlation exceeds every shuffled result for both models (one-sided $p \approx 0.00020$, using the plus-one correction;[23] SI Note 7). Furthermore, among all three target pairs whose IQRs differed by at most 10%, the smoother target also had the lower mean normalized test error for both SchNet and GemNet (SI Note 7).

Together, these results support architecture-aligned target roughness (D_{IQR}) as a scale-invariant candidate pre-training diagnostic of relative target learnability. Molecular-neighbourhood alignment contributes information beyond the shuffled target-distribution baseline, although normalization contributes substantially to the observed association. These retrospective results do not establish a causal mechanism or guarantee predictive performance for new target families.

2.5 Target design controls out-of-domain transfer

To evaluate whether favourable target design enhances transferability beyond the training distribution, we tested model predictions on the QM9 dataset [16]. While QM9 shares elements with tmQM, it shifts the chemical domain from transition-metal complexes to small organic molecules, altering molecular size, bonding motifs, and coordination environments. To isolate this chemical domain shift from discrepancies in electronic-structure fidelity, QM9 DFT reference energies were recomputed at the identical level of theory used for tmQM, with transformed targets generated using frozen, train-only baseline parameters (SI Methods).

Evaluating out-of-domain (OOD) performance across distinct chemical families can introduce systematic dataset-level energy offsets due to divergent stoichiometry and size distributions. Because raw OOD MAE combines these global offsets with molecule-to-molecule errors, we assessed trend preservation using two complementary metrics: Kendall rank correlation (τ), which quantifies energetic ordering, and mean-shifted MAE, which measures residual scatter after removing a single dataset-wide offset (SI Note 5). This offset is estimated from the QM9 reference labels, rendering mean-shifted MAE an oracle diagnostic rather than an independently calibrated prediction score. All reconstructed routes were mapped back onto the parent DFT-BoE energy target prior to comparison.

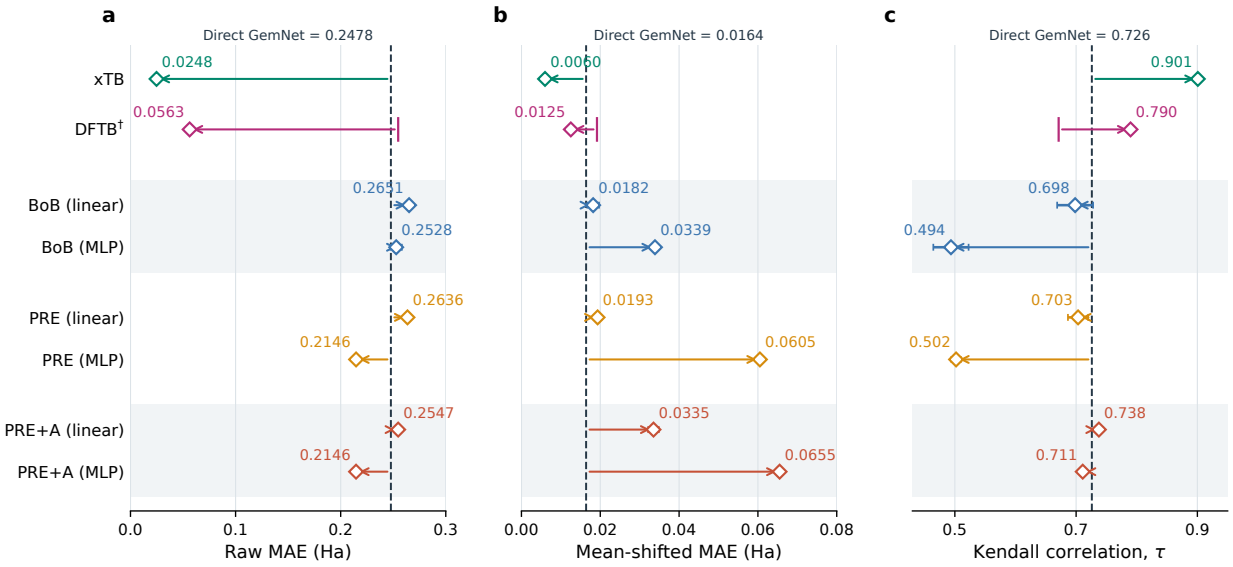


Figure 4: **Out-of-domain GemNet transfer from tmQM to QM9. a,** Raw MAE. **b,** Mean-shifted MAE using one offset estimated from QM9 labels. **c,** Kendall correlation. Diamonds show baseline-plus-GemNet predictions against $E_{\mathrm{ref,BoE}}$; dashed lines mark direct GemNet. Primary results are three-seed means with sample-SD bars on 128,988 common molecules. Magenta DFTB† is a seed-101 control on 128,978 valid molecules; its short reference ticks and arrows use direct seed 101 on the same subset (SI Notes 5 and 8).

Among all primary GemNet routes evaluated on QM9, the composition-referenced xTB residual achieved the best mean transfer performance across the three reported metrics (Fig. 4). Learning on the xTB residual reduced mean-shifted MAE from 0.01641 Ha for direct GemNet to 0.00598 Ha while increasing Kendall τ from 0.726 to 0.901 (Fig. 4b,c). It also yielded the lowest raw MAE, reducing error from 0.2478 to 0.0248 Ha (Fig. 4a), showing that its transfer advantage does not rely on mean-shift offset calibration. In contrast, no local descriptor baseline improved upon direct GemNet on both trend metrics; linear PRE+A slightly increased mean Kendall correlation ($\tau = 0.738$) but increased mean-shifted MAE to 0.03351 Ha (Fig. 4b,c). Furthermore, GemNet preserved the ordering of the predicted xTB residual itself ($\tau = 0.789$ on the common QM9 cohort), supporting transfer of the learned residual correction rather than only the reconstructed energy ranking (SI Note 5). SchNet exhibited parallel improvements on the xTB residual, reducing mean-shifted MAE from 0.01563 to 0.00665 Ha and increasing Kendall τ from 0.746 to 0.895 (SI Note 5).

Semiempirical density-functional tight-binding (DFTB†) provided supporting evidence on its valid subset, reducing raw MAE from 0.2547 to 0.0563 Ha, reducing mean-shifted MAE from 0.01917 to 0.01253 Ha, and increasing Kendall τ from 0.671 to 0.790 relative to matched-subset direct GemNet (SI Notes 5 and 8). This seed-101 control uses a different training subset and does not provide a three-seed uncertainty estimate.

These results motivate the principle of **baseline complementarity** in physical target design. Baselines constructed from local geometric descriptors share information with message-passing GNNs and may remove smooth variations that message-passing layers naturally learn while leaving behind residuals that are rough relative to their scale. In contrast, semiempirical quantum methods supply molecule-wide electronic-structure priors—including interactions beyond standard spatial cutoffs—that may complement downstream neural architectures. Tu and Rowley demonstrated such complementarity for long-range interactions, showing that a DFTB3 baseline supplies intermolecular

interactions beyond a local neural network's cutoff and improves transferability over direct learning [24]. While target scale and representation smoothness provide complementary diagnostics rather than strict guarantees for transfer, combining semiempirical physical baselines with architecture-aligned GNNs improves out-of-domain generalization in the present comparisons; the specific physical mechanism remains unresolved.

3 Discussion

In this study, we demonstrate that Δ-learning target transformations change two distinct properties: target scale (IQR) and representation-space roughness relative to the downstream architecture. While target shrinkage proportionally lowers absolute prediction error at fixed normalized error, a disproportionate increase in scale-normalized roughness (D_{IQR}) is associated with poorer scale-normalized downstream performance. Distinguishing these two factors establishes a two-dimensional target-design framework for interpreting Δ-learning, providing a rational principle for assessing baselines beyond their accuracy alone.

Evaluating a hierarchy of baseline models reveals that baseline accuracy and residual scale alone cannot predict downstream GNN performance. Local descriptor baselines (Bag-of-Bonds, PRE, and PRE+A) share geometric information with local message-passing GNNs and leave smaller residuals that are disproportionately rough relative to their scale. Their MLP residuals exhibit poorer test performance than direct DFT-BoE learning, consistent with a generalization limitation associated with residual roughness, without ruling out optimization or network capacity effects. In contrast, the xTB baseline reduces both target scale and normalized roughness while improving test performance. DFTB also improves matched-subset test MAE, but slightly increases normalized roughness, showing that roughness reduction is not necessary for lower absolute error.

The success of semiempirical baselines motivates the principle of **baseline complementarity**. Baselines that mirror the inductive biases of the downstream architecture may remove target variations that message-passing layers already represent efficiently, leaving behind rougher, harder-to-learn residuals. Effective baselines may instead complement the downstream model by absorbing energy contributions that local architectures represent poorly, enabling the GNN to learn a scale-reduced target with favourable residual structure. This is a physical interpretation of the observed trends, not an identified causal mechanism.

Because post-training evaluation metrics require full model optimization, pre-training diagnostics are useful for practical target engineering. Scale-normalized graph Dirichlet roughness (D_{IQR}) measures target variation over architecture-aligned representation proxies without training a downstream GNN. Its strong semi-log association with normalized test error ($n-$MAE) supports D_{IQR} as a candidate diagnostic for screening baselines before committing to computationally expensive model training. Shared normalization contributes substantially to this association, while joint-label permutation controls support an additional contribution from molecular-neighbourhood alignment. The present test-set analysis is retrospective; prospective screening requires an independent evaluation set.

Finally, out-of-domain transfer to small organic molecules (QM9) shows that favourable target design can improve generalization beyond the training domain. The xTB residual achieves superior trend preservation and energetic rank correlation compared to local descriptor baselines, with DFTB providing supporting evidence on its valid subset. These results support the transfer value of semiempirical baselines across distinct chemical environments, without establishing that roughness alone predicts transfer or that the learned molecular representations have improved.

4 Conclusion

Target transformations in scientific machine learning cannot be evaluated solely by scale reduction or baseline accuracy; residual scale and scale-normalized representation smoothness provide complementary diagnostics of their success. By formalizing this interplay into a target-design framework that jointly considers scale and smoothness—and supporting D_{IQR} as a candidate pre-training diagnostic—this work transforms target preparation from an ad hoc preprocessing step into a principled design paradigm. Just as neural networks are engineered to respect physical symmetries, baseline transformations should be chosen with the downstream representation space in mind. Extending this framework to diverse physical properties, multi-fidelity workflows, and machine learning interatomic potentials offers a foundation for building generalizable, transferable models across the physical sciences.

5 Methods

5.1 Datasets and targets

The in-domain data are the aligned neutral tmQM transition-metal complexes with ωB97M-V/def2-SVPD energies and fixed train, validation, and test molecule identifiers [15, 25, 26]. The full panel contains 56,830 training, 7,104 validation, and 7,104 test molecules. The six descriptor residuals use 56,821 training and 7,103 validation molecules, while all 13 primary targets share the same 7,104 test molecules. Both architectures use identical target values and split memberships within each target. DFTB success filtering leaves 56,540 training, 7,068 validation, and 7,068 test molecules; failed calculations were not imputed. QM9 is used only for OOD evaluation; its energies were recomputed at the same level with Psi4 to avoid conflating chemical-domain shift with electronic-structure mismatch [16, 27].

For method m, molecular energies were first corrected using fixed element-wise atomic references,

$$E^{m}_{\mathrm{ref}}(i) = E^{m}_{\mathrm{raw}}(i) - \sum_{Z} n_Z(i)\epsilon^{m}_{Z}, \tag{6}$$

then corrected by a linear model of elemental counts fitted only on the tmQM training split to obtain $E^{m}_{\mathrm{ref,BoE}}$. Atomic references are fixed external inputs, not parameters fitted to held-out molecular energies. Separate composition models are fitted for DFT, xTB, and DFTB and frozen for validation, test, and QM9 evaluation. The DFT quantity is the parent target. Delta targets subtract either a fitted descriptor baseline or the consistently referenced xTB/DFTB energy. Full target-generation, masking, and molecule-ID alignment details are in SI Methods.

5.2 Baseline models

BoE denotes elemental-count features and the standard BoE reference is linear. A composition-only control fits a nonlinear MLP to the same counts and atom-referenced DFT energy, thereby changing baseline flexibility without adding structural information. BoB denotes the study-specific RDKit-derived typed-bond-count vector, not the conventional Coulomb-matrix Bag-of-Bonds descriptor. PRE expands local element-resolved pair distances in a radial basis, and PRE+A augments PRE with angular three-body terms. BoB, PRE, and PRE+A each have linear ridge and MLP fits to the DFT-BoE parent target; all four MLP baselines share hidden widths of 512 and 128 and the same training protocol. Ridge regularization was selected by five-fold training-only cross-validation, with preprocessing refitted within each fold. Final baseline parameters and preprocessing were fitted

using only the corresponding tmQM training split. The validation split was used to select each MLP checkpoint, while test labels were used only for final evaluation. Baseline fitting seed 101 defines the canonical residual targets; fitting-seed sensitivity is reported separately in SI Note 9. GFN2-xTB and DFTB are fixed semiempirical baselines [19, 20, 21]. DFTB uses SCC-DFTB/DFTB2 with the PTBP Slater–Koster set, without third-order or dispersion corrections [28], and only paired successful calculations; failures were not imputed. Feature dimensions, hyperparameters, and failure audits are in SI Methods and SI Note 8.

5.3 GNN training

GemNet-T is the primary downstream model and SchNet is the architecture control [14, 4]. Models were trained on energies only, without forces or periodic boundary conditions, using fixed splits, a 6.0 Å cutoff, and a common FairChem/OCP-style protocol. Every target was standardized by its own training-set mean and population standard deviation. Within each architecture, architecture settings, loss, AdamW configuration, initial learning rate, scheduler rule, batch size, and checkpoint criterion were held fixed across targets. AdamW with AMSGrad used an initial learning rate of 5×10^{-4}, MSE energy loss, and an 80-epoch budget; batch sizes were 16 for SchNet and four for GemNet. ReduceLROnPlateau responded independently to each validation trajectory, with factor 0.8 and patience three, producing target-specific learning-rate schedules. Each run's best-validation checkpoint was evaluated on both the tmQM test set and QM9. All 13 primary targets use independent GNN seeds 101, 102, and 103 for both architectures; the two DFTB targets use seed 101 only for both architectures, giving 82 runs in total. Reported primary results are means of seed-level metrics, not ensemble predictions, with sample standard deviations shown where applicable. Single-seed DFTB results have no across-seed uncertainty estimate. Exact seed-level results are reported in SI Note 9. Complete configurations and training diagnostics are in SI Methods.

5.4 Roughness and transfer diagnostics

GemNet roughness uses standardized PRE+A features and SchNet roughness uses standardized PRE features. A cosine $k = 10$ nearest-neighbour graph was built on the 7,104-molecule tmQM test cohort. Zero-variance feature channels were removed, and retained channels were divided by the graph cohort's population standard deviation without centring. This unsupervised graph preprocessing is distinct from train-only baseline preprocessing. Unique undirected union edges were assigned similarity-based weights using the locally scaled Gaussian kernel in Equation 2, and D_{IQR} was computed as defined in Section 2.2. Both D_{IQR} and n-$\mathrm{MAE}_{\mathrm{test}}$ use the corresponding test-target IQR; graphs and weights remain fixed across primary targets within each architecture. DFTB controls use separately constructed valid-subset test graphs and are excluded from primary regressions and pooled statistics.

Semi-log regressions use unweighted ordinary least squares across the 13 primary target means, with an intercept and $\ln D_{\mathrm{IQR}}$ as predictor. Raw-error, raw-roughness, variance-normalization, partial-correlation, and leave-one-target-out sensitivity analyses are reported in SI Note 7. The same note reports 5,000 joint test-label permutations, which preserve each target's distribution and IQR while breaking molecular-neighbourhood alignment, and comparisons of target pairs whose IQRs differ by at most 10%. Leave-one-target-out fits are influence checks, not cross-validated prediction scores. These test-set analyses are retrospective diagnostics performed after model selection; test labels were not used to fit baselines, schedule training, or select checkpoints.

5.5 OOD transfer evaluation

tmQM-trained models were evaluated without retraining on QM9 geometries, using frozen tmQM training-only baseline parameters. Residual predictions were combined with their corresponding baselines and mapped to the common DFT-BoE parent before full-target correlation and ranking metrics were calculated. Primary parent-energy comparisons use 128,988 common QM9 molecules and three GNN seeds. The DFTB residual is shown on its 128,978-molecule intersection with this cohort, alongside a matched-subset direct seed-101 reference; this matches evaluation molecules, not training cohorts. Mean-shifted MAE removes one scalar mean-error offset estimated separately for each predictor from the evaluated QM9 reference labels and is used only as a post hoc oracle diagnostic, not an independently calibrated prediction score. Figure 4 reports GemNet raw and mean-shifted MAEs in Ha, without IQR normalization, and Kendall correlations; exact coverage, offset magnitudes, and additional metrics are reported in SI Note 5. Detailed reconstruction and evaluation procedures are in SI Methods.

Data and code availability

The code, processed target tables, split identifiers, evaluation summaries, figure source data, manuscript sources, and reproducibility checks for this work are available in the corrected release `v0.2.0-corrected` of the project repository [29]. The original tmQM and QM9 datasets are available from their respective providers [15, 16]. Raw structures, LMDB databases, model checkpoints, and full prediction arrays are not redistributed in this release; the included manifests document their provenance.

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Author contributions

K.M.G. conceived and designed the study, performed the computational experiments, analysed the data, prepared the figures, and wrote the original manuscript. I.N. contributed to data preparation, computational workflows, and interpretation. O.V. and S.H. supervised the project. O.V. contributed to study design, interpretation, and manuscript revision. All authors reviewed and approved the manuscript.

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Supplementary Information for The Mechanics of Delta Learning: Target Design for Generalizable Scientific Machine Learning

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Scope and Reporting Conventions

This supplement asks how target construction changes the difficulty of learning molecular energies and their transfer between chemical domains. SchNet and GemNet are evaluated on the same 13 target definitions, with independent GNN seeds 101, 102, and 103 for every target. Each run is evaluated on the tmQM test split and on QM9 using its own best-validation checkpoint. The 78 main runs are supplemented by four DFTB runs: two targets, two architectures, and seed 101 only.

Unless stated otherwise, errors are in Hartree (Ha), and uncertainty denotes the sample standard deviation of independently evaluated seed-level scores. Predictions are not averaged into an ensemble. A smaller error measures better performance on that target; it does not by itself explain the physical origin of the improvement.

DFTB is shown in magenta and marked † in every comparison figure. It is a supporting control, not a member of the primary target panel. DFTB is excluded from all pooled correlations, regression fits, target-ranking statistics, leave-one-target-out analyses, and primary-panel summaries. Its own single-run metrics and explicitly matched-subset comparisons are reported separately. No across-seed error bar is assigned to a single DFTB run.

The target-panel table defines the identifiers used in plots. BoE-lin means the linear composition reference; BoE-MLP denotes the alternative nonlinear composition residual. An unsubscripted BoE reference is linear. In particular, the BoE-MLP target is fitted directly from $E_{\mathrm{ref}}^{\mathrm{DFT}}$, not from the linear-BoE residual.

Table S1: Complete target and evaluation panel. Counts are molecules per seed; both architectures use the same counts. DFTB† rows are display-only controls for pooled analyses.

ID	Target	Seeds	Train	Validation	Test	QM9
T1	$E_{\mathrm{ref}}^{\mathrm{DFT}}$	3	56,830	7,104	7,104	130,807
T2	$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	3	56,830	7,104	7,104	130,807
T3	$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	3	56,830	7,104	7,104	130,807
T4	$E_{\mathrm{ref}}^{\mathrm{xTB}}$	3	56,830	7,104	7,104	130,807
T5	$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	3	56,830	7,104	7,104	130,807
T6	$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	3	56,830	7,104	7,104	130,807
T7	$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	3	56,830	7,104	7,104	130,807
T8	$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	3	56,821	7,103	7,104	130,807
T9	$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	3	56,821	7,103	7,104	130,807
T10	$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	3	56,821	7,103	7,104	128,988
T11	$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	3	56,821	7,103	7,104	128,988
T12	$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	3	56,821	7,103	7,104	128,988
T13	$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	3	56,821	7,103	7,104	128,988
T14†	$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	1	56,540	7,068	7,068	130,797
T15†	$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	1	56,540	7,068	7,068	130,797

The architecture comparison is matched within every target: both architectures use its identical target values and split memberships. The entire panel does not have an identical training cohort across targets. The six descriptor residuals use 56,821 training and 7,103 validation molecules, whereas the other main targets use 56,830 and 7,104. All 13 main targets share the same 7,104 test molecules. DFTB has a separate success-filtered cohort.

Supplementary Section 1: Target Preprocessing and Scale Controls

Supplementary Note 1: Direct Preprocessing Controls

Composition referencing removes a large, learnable contribution to the atom-referenced DFT energy. With a linear BoE reference, test MAE falls from 0.019917 ± 0.000178 to 0.015648 ± 0.000077 Ha for SchNet and from 0.013360 ± 0.000423 to 0.009448 ± 0.000469 Ha for GemNet. The nonlinear BoE baseline narrows the residual further, but downstream errors increase to 0.024672 ± 0.000112 and 0.022380 ± 0.000452 Ha, respectively. A smaller residual distribution therefore does not guarantee an easier regression problem.

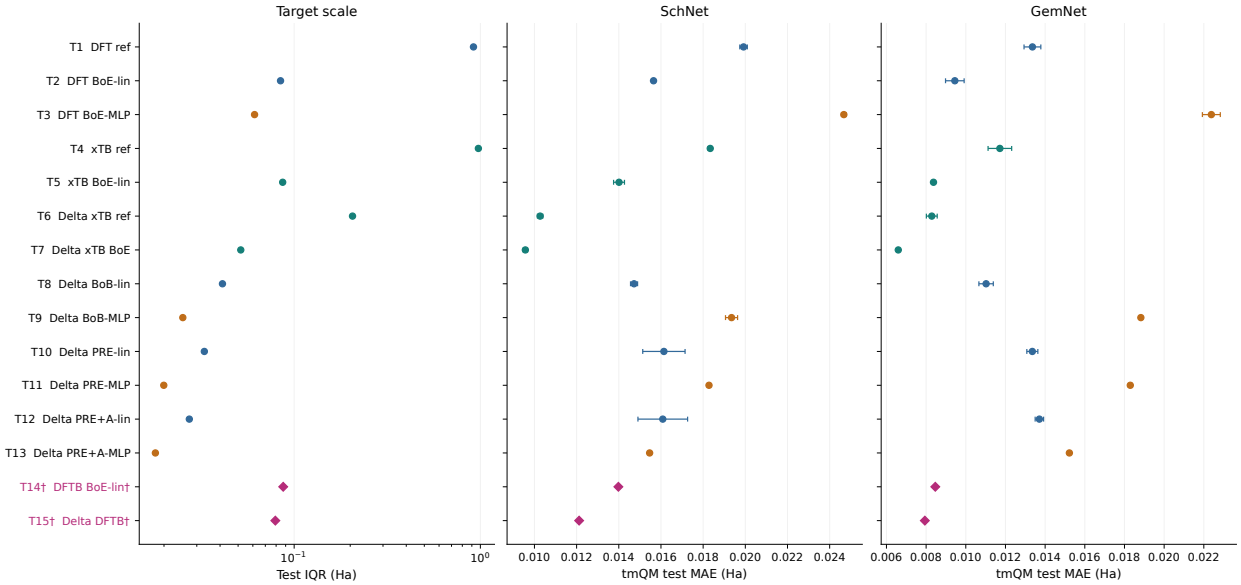


Figure S1: **Target scale and test error.** Test IQR and tmQM test MAE for all targets. Error bars are sample SD over three GNN seeds. Magenta DFTB† points are single-seed, reduced-cohort controls.

Table S2: Native-target test MAE (Ha), mean ± sample SD across seeds 101, 102, and 103. DFTB† has seed 101 only. xTB and DFTB direct-energy rows measure their own energies, not DFT reconstruction error.

Target	SchNet test MAE	GemNet test MAE
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	0.019917 ± 0.000178	0.013360 ± 0.000423
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.015648 ± 0.000077	0.009448 ± 0.000469
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.024672 ± 0.000112	0.022380 ± 0.000452
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.018337 ± 0.000099	0.011718 ± 0.000595
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	0.014011 ± 0.000258	0.008372 ± 0.000031
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.010269 ± 0.000128	0.008286 ± 0.000273
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.009569 ± 0.000101	0.006596 ± 0.000048
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.014724 ± 0.000167	0.011024 ± 0.000363
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.019347 ± 0.000287	0.018826 ± 0.000056
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.016139 ± 0.001004	0.013355 ± 0.000278
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.018278 ± 0.000070	0.018293 ± 0.000090
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.016086 ± 0.001178	0.013710 ± 0.000212
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.015462 ± 0.000035	0.015222 ± 0.000006
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	0.013980	0.008462
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.012119	0.007936

Direct xTB and DFTB energy-prediction rows describe how well each GNN learns the semiempirical energy itself. They are not DFT reconstruction errors. For a residual $\Delta^b = y - b$, in contrast, the stacked predictor $\hat{y} = b + \widehat{\Delta}^b$ satisfies

$$\hat{y} - y = \widehat{\Delta}^b - \Delta^b. \tag{S1}$$

Residual MAE and RMSE are exactly those of the reconstructed parent-energy prediction on the same molecules. This equivalence does not extend to R^2, Pearson correlation, or rank correlation without reconstructing the parent energy.

Supplementary Note 2: What Scale Normalization Removes

For $y' = ay + c$, with $a \neq 0$, the IQR and a consistently rescaled prediction error change by $|a|$. For a fixed molecular graph,

$$\mathrm{IQR}(y') = |a|\mathrm{IQR}(y), \quad D_{\mathrm{raw}}(y') = a^2 D_{\mathrm{raw}}(y), \quad D_{\mathrm{IQR}}(y') = D_{\mathrm{IQR}}(y). \tag{S2}$$

IQR-normalized roughness therefore removes a pure change of units or amplitude. A fitted baseline is not generally a uniform rescaling: it can change molecular ordering, tail structure, and local differences.

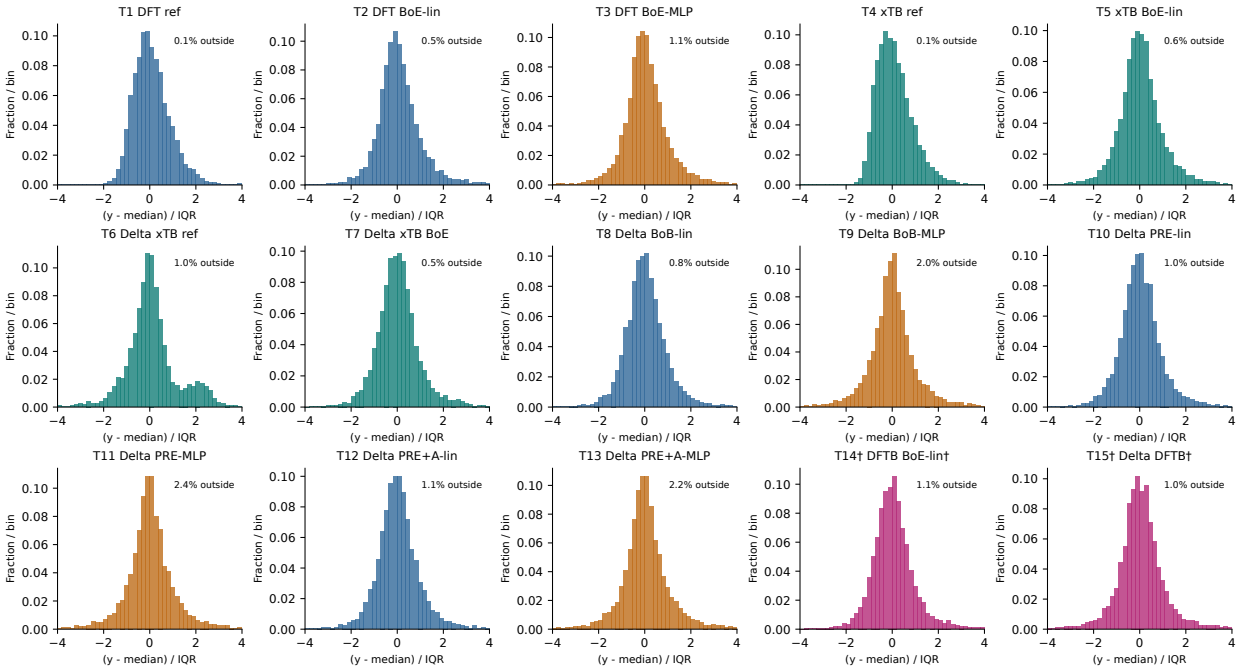


Figure S2: **Standardized test-target distributions.** Targets are centred by their test median and divided by their test IQR. Bin heights use the full test target population; annotations give fractions outside the displayed interval. DFTB† uses its valid subset.

This invariance is algebraic; it does not establish that independently optimized networks attain identical normalized errors. Nor does it eliminate correlations introduced by using a related scale factor on both diagnostic axes. Supplementary Note 7 examines that issue directly.

Supplementary Section 2: Baseline-Selection Diagnostics

Supplementary Note 3: Baseline Accuracy and Downstream Error

A useful baseline must do more than reproduce the parent energy accurately: the remaining signal must also be learnable by the downstream architecture. The baseline-score table reports frozen-baseline errors on test molecules. Test scores were calculated after fitting and did not select baseline parameters.

Table S3: Frozen baseline performance against the DFT-BoE parent (Ha). Errors and correlations refer to the baseline alone; test labels are used here only for post-training evaluation. DFTB† uses its valid subset.

Residual	Test MAE	Test RMSE	Pearson r	Spearman ρ
$\Delta^{\mathrm{BoB}}_{\mathrm{lin}}$	0.029315	0.047333	0.839	0.823
$\Delta^{\mathrm{BoB}}_{\mathrm{MLP}}$	0.021124	0.041346	0.881	0.901
$\Delta^{\mathrm{PRE}}_{\mathrm{lin}}$	0.024967	0.057663	0.771	0.893
$\Delta^{\mathrm{PRE}}_{\mathrm{MLP}}$	0.018746	0.068647	0.731	0.934
$\Delta^{\mathrm{PRE+A}}_{\mathrm{lin}}$	0.021128	0.047002	0.847	0.923
$\Delta^{\mathrm{PRE+A}}_{\mathrm{MLP}}$	0.016335	0.053602	0.811	0.950
$\Delta^{\mathrm{xTB}}_{\mathrm{ref,BoE}}$	0.035716	0.050873	0.830	0.787
$\Delta^{\mathrm{DFTB}\dagger}_{\mathrm{ref,BoE}}$	0.058773	0.086576	0.560	0.504

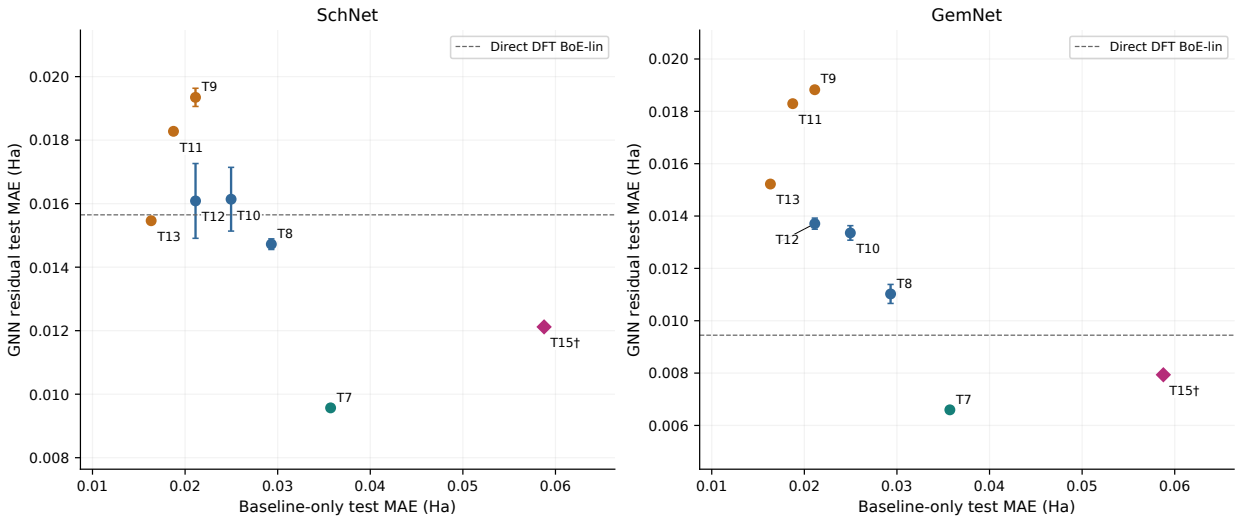


Figure S3: **Baseline accuracy versus downstream test error.** Each point pairs baseline-only test MAE with the GNN residual test MAE. Dashed lines show direct DFT-BoE learning. Error bars are GNN seed SD; DFTB† is a separate single-seed control.

More accurate descriptor baselines need not yield better downstream predictors. BoB, PRE, and PRE+A MLPs have smaller baseline errors than their corresponding linear models, yet none of their residual targets beats direct DFT-BoE learning for GemNet. For SchNet, linear BoB improves on direct learning, whereas linear PRE and PRE+A do not improve the mean test MAE. The latter two targets also have larger seed variability than most other SchNet targets.

The xTB BoE-referenced residual has the smallest mean tmQM test MAE in the 13-target panel for both architectures: 0.009569 ± 0.000101 Ha for SchNet and 0.006596 ± 0.000048 Ha for GemNet. Relative to direct DFT-BoE learning, these are 1.635-fold and 1.432-fold lower errors. This is an empirical target-design result, not proof of a particular electronic or geometric mechanism.

Supplementary Note 4: Descriptor Baseline Capacity

The linear and MLP descriptor models have the same parent target and, within each descriptor family, the same inputs and split memberships. Changing the fitting model tests baseline capacity without changing the DFT reference convention. Comparing descriptors under the common 512–128-hidden-unit MLP design tests information content, although input dimensions and total parameter counts differ.

MLP-PRE and MLP-PRE+A concentrate much of the residual near zero, but downstream errors and roughness do not decrease in proportion to the IQR. Their in-domain performance should not be extrapolated to QM9: fixed descriptor baselines can behave differently outside the tmQM domain. The transfer evaluation below tests this directly.

Supplementary Section 3: QM9 Transfer Controls

Supplementary Note 5: Complete Transfer Evaluation

Every main GNN seed is evaluated on QM9 without refitting the GNN or baseline. Native-target MAE is first reported on all molecules for which the target is defined: 130,807 for most targets, 128,988 for PRE and PRE+A residuals, and 130,797 for DFTB. Missing descriptor and DFTB rows are not imputed. Because cohorts differ, native-score tables document coverage but are not exact all-target, matched-cohort comparisons.

A second analysis reconstructs the common DFT-BoE parent and evaluates all main DFT-prediction routes on the same 128,988 molecules. Direct xTB and DFTB energy predictors are not DFT-prediction routes and do not enter these parent-energy tables. The displayed DFTB residual uses its 128,978-molecule intersection with this cohort. An additional exact-mask comparison with direct DFT and xTB appears in Supplementary Note 8.

For each seed, the oracle mean-shift diagnostic subtracts the mean prediction error on the evaluated QM9 cohort,

$$\hat{y}_i^{\text{shifted}} = \hat{y}_i - \frac{1}{N}\sum_{j=1}^{N}(\hat{y}_j - y_j). \tag{S3}$$

This uses QM9 reference labels and is not a deployable, uncalibrated prediction. It removes constant bias, but not ordering errors, heteroscedastic scatter, or outliers. It minimizes squared error over constant shifts, not necessarily MAE. Raw transfer errors remain essential.

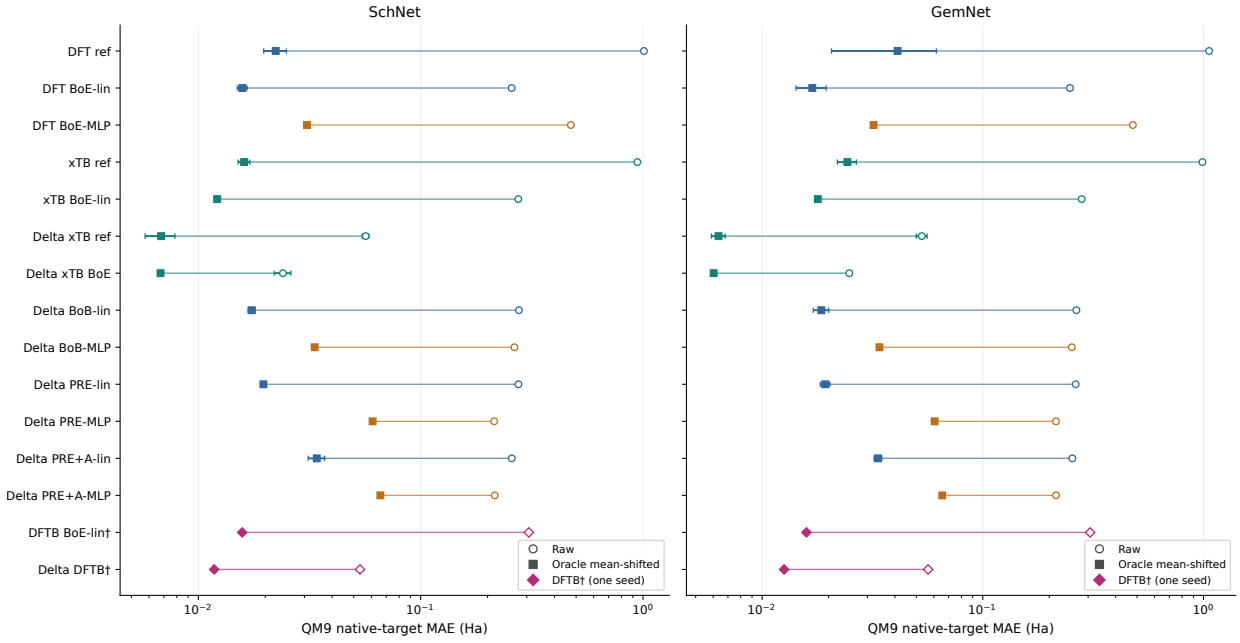


Figure S4: **Raw and oracle-shifted QM9 errors.** Open and filled symbols show native-target raw and shifted MAE; lines join the two diagnostics. Main results are three-seed means with sample SD. DFTB† is shown separately in magenta.

Table S4: SchNet QM9 native-target MAEs (Ha). Main targets: three independent seed scores, mean ± sample SD; DFTB†: seed 101. Native cohorts differ as listed. Shifts are estimated from the evaluated QM9 labels, not a separate calibration set.

Target	N	Raw MAE	Oracle-shift MAE
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	130,807	1.009072 ± 0.005934	0.022291 ± 0.002609
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	130,807	0.256637 ± 0.004093	0.015778 ± 0.000802
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	130,807	0.473662 ± 0.009143	0.030793 ± 0.000568
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	130,807	0.942905 ± 0.001125	0.016076 ± 0.000972
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	130,807	0.274700 ± 0.002912	0.012149 ± 0.000225
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	130,807	0.056567 ± 0.001925	0.006805 ± 0.001043
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	130,807	0.024017 ± 0.002079	0.006760 ± 0.000192
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	130,807	0.276571 ± 0.002836	0.017383 ± 0.000645
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	130,807	0.264029 ± 0.002472	0.033360 ± 0.000369
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	128,988	0.275298 ± 0.002464	0.019612 ± 0.000547
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	128,988	0.214158 ± 0.000980	0.060797 ± 0.000118
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	128,988	0.256813 ± 0.001379	0.034096 ± 0.002906
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	128,988	0.215363 ± 0.000066	0.065843 ± 0.000111
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	130,797	0.306405	0.015751
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	130,797	0.053349	0.011783

Table S5: GemNet QM9 native-target MAEs (Ha). Main targets: three independent seed scores, mean ± sample SD; DFTB†: seed 101. Native cohorts differ as listed. Shifts are estimated from the evaluated QM9 labels, not a separate calibration set.

Target	N	Raw MAE	Oracle-shift MAE
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	130,807	1.060215 ± 0.018798	0.041115 ± 0.020489
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	130,807	0.248256 ± 0.006282	0.016878 ± 0.002639
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	130,807	0.478472 ± 0.004472	0.031996 ± 0.000350
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	130,807	0.988524 ± 0.003253	0.024332 ± 0.002402
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	130,807	0.280595 ± 0.003342	0.017877 ± 0.000265
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	130,807	0.052928 ± 0.002964	0.006344 ± 0.000454
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	130,807	0.024815 ± 0.000231	0.006029 ± 0.000117
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	130,807	0.265518 ± 0.002632	0.018540 ± 0.001494
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	130,807	0.253024 ± 0.005363	0.034020 ± 0.000691
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	128,988	0.263566 ± 0.000964	0.019325 ± 0.000974
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	128,988	0.214647 ± 0.000882	0.060508 ± 0.000347
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	128,988	0.254705 ± 0.002866	0.033508 ± 0.001345
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	128,988	0.214623 ± 0.000429	0.065508 ± 0.000444
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	130,797	0.306466	0.015873
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	130,797	0.056495	0.012578

Parent-Energy Reconstruction and Trend Fidelity

Equation S1 allows raw MAE to be scored in residual space. Parent-energy R^2 and Kendall τ require reconstruction first. For E_{ref} and the nonlinear BoE route, the frozen linear DFT-BoE term is subtracted after reconstruction so that every route refers to the same parent quantity.

Table S6: SchNet transfer after reconstruction onto the common DFT-BoE parent. Primary routes use the identical 128,988-molecule descriptor-common QM9 cohort; DFTB† uses its 128,978-molecule intersection. RMSE and R^2 are means of seed-level oracle-shifted metrics. Kendall ordering is shift-invariant.

Route	Oracle MAE	RMSE	R^2	Kendall τ
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	0.020707 ± 0.002107	0.037080	0.467	0.749 ± 0.008
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.015634 ± 0.000814	0.028811	0.681	0.746 ± 0.015
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.030699 ± 0.000419	0.043151	0.285	0.549 ± 0.034
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.006603 ± 0.001017	0.009128	0.967	0.898 ± 0.013
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.006647 ± 0.000190	0.009381	0.966	0.895 ± 0.001
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.017229 ± 0.000583	0.030164	0.650	0.718 ± 0.012
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.033290 ± 0.000453	0.045189	0.216	0.473 ± 0.009
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.019612 ± 0.000547	0.032431	0.596	0.687 ± 0.002
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.060797 ± 0.000118	0.108187	-3.495	0.506 ± 0.006
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.034096 ± 0.002906	0.066634	-0.709	0.734 ± 0.002
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.065843 ± 0.000111	0.117543	-4.306	0.710 ± 0.000
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.011623	0.016650	0.893	0.810

Table S7: GemNet transfer after reconstruction onto the common DFT-BoE parent. Primary routes use the identical 128,988-molecule descriptor-common QM9 cohort; DFTB† uses its 128,978-molecule intersection. RMSE and R^2 are means of seed-level oracle-shifted metrics. Kendall ordering is shift-invariant.

Route	Oracle MAE	RMSE	R^2	Kendall τ
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	0.039317 ± 0.018274	0.062145	-0.697	0.651 ± 0.029
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.016407 ± 0.002413	0.030401	0.643	0.726 ± 0.048
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.031678 ± 0.000363	0.044438	0.242	0.547 ± 0.024
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.006311 ± 0.000461	0.008690	0.971	0.896 ± 0.009
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.005982 ± 0.000095	0.008393	0.973	0.901 ± 0.001
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.018193 ± 0.001493	0.032728	0.587	0.698 ± 0.030
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.033881 ± 0.000583	0.045824	0.193	0.494 ± 0.029
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.019325 ± 0.000974	0.032730	0.588	0.703 ± 0.017
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.060508 ± 0.000347	0.107942	-3.475	0.502 ± 0.008
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.033508 ± 0.001345	0.066759	-0.712	0.738 ± 0.007
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.065508 ± 0.000444	0.117164	-4.272	0.711 ± 0.001
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.012526	0.018161	0.873	0.790

On the matched main cohort, both xTB residual formulations retain substantially smaller oracle-shifted errors than the descriptor-MLP residuals. The BoE-referenced xTB route gives approximately 0.00665 Ha for SchNet and 0.00598 Ha for GemNet, with Kendall τ near 0.895 and

0.901. The atom-referenced xTB residual is also competitive; composition referencing does not uniformly improve every transfer metric for both architectures.

MLP-PRE and MLP-PRE+A have oracle-shifted errors near 0.061–0.066 Ha and negative parent-energy R^2, despite small in-domain residual widths. PRE+A can retain a moderate Kendall ordering while having poor RMSE and R^2: rank fidelity does not protect against large amplitude errors or outliers. These failures limit claims that a flexible fitted baseline necessarily improves transfer.

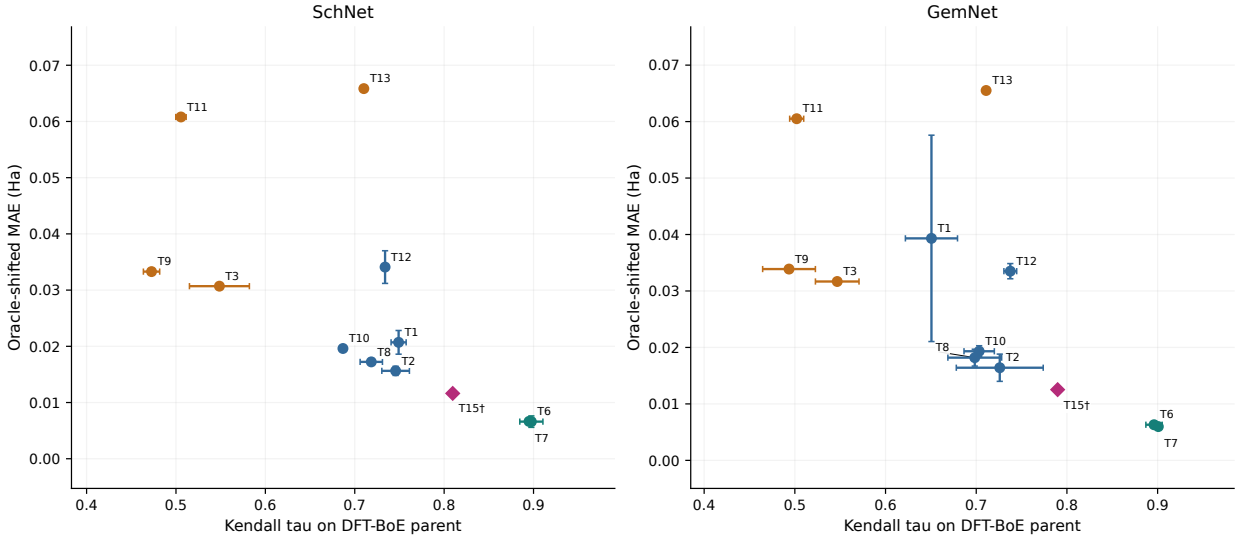


Figure S5: **Trend fidelity and remaining QM9 error.** Kendall τ versus oracle-shifted MAE after DFT-BoE reconstruction. Main routes use 128,988 common molecules; DFTB† uses its 128,978-molecule intersection and is not pooled with the main routes.

Supplementary Section 4: Architecture Controls and Roughness

Supplementary Note 6: Matched Architecture Panels

The roughness analysis uses all 13 main targets for both architectures. SchNet's proxy is PRE, and GemNet's proxy is PRE+A. These are fixed geometric proxies, not learned network representations. Pairwise message passing can encode more complex environments through multiple layers; PRE is not an exact representation of everything SchNet can learn.

The primary graphs use the 7,104 test molecules shared by all 13 targets. Standardized descriptors define cosine $k = 10$ neighbours. Unique undirected edges receive locally scaled Gaussian weights. For target y,

$$D_{\mathrm{raw}}(y) = \frac{\sum_{\{i,j\}\in\mathcal{E}} w_{ij}(y_i - y_j)^2}{\sum_{\{i,j\}\in\mathcal{E}} w_{ij}}, \qquad D_{\mathrm{IQR}}(y) = \frac{D_{\mathrm{raw}}(y)}{\mathrm{IQR}_{\mathrm{test}}(y)^2}. \tag{S4}$$

The normalized response is $n\text{-MAE}_{\mathrm{test}} = \mathrm{MAE}_{\mathrm{test}}/\mathrm{IQR}_{\mathrm{test}}(y)$. Test error, its scale denominator, and roughness use the same test cohort. Molecular descriptors alone define graph edges and weights; test labels supply the target differences and IQR after models are frozen. Validation is used for model selection and learning-rate scheduling, not the reported performance or roughness. This is a retrospective cross-target diagnostic, not a prospectively validated model-selection rule; using these test results to select future models would require a new independent evaluation set.

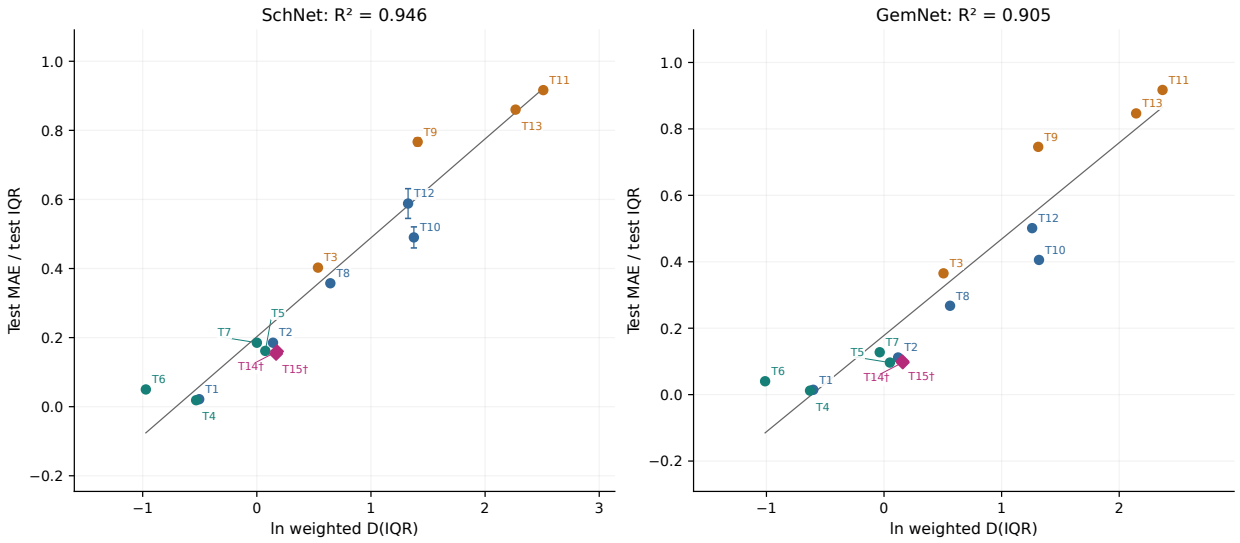


Figure S6: **Weighted test roughness versus normalized test error.** Lines fit the 13 main target means only. Error bars are three-seed SD divided by the fixed test IQR. DFTB† uses valid-subset test graphs and is excluded from fits and R^2.

Table S8: Architecture-proxy roughness and normalized test errors. Primary graph: 7,104 common test molecules; DFTB† graph: 7,068 DFTB-valid test molecules. Each error and roughness uses its matching test IQR. DFTB positions are visual controls, not members of the regression.

Target	Graph IQR (Ha)	PRE D_{IQR}	PRE+A D_{IQR}	SchNet n-MAE	GemNet n-MAE
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	0.918321	0.6036	0.5477	0.0217	0.0145
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.084504	1.1522	1.1271	0.1852	0.1118
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.061288	1.7086	1.6583	0.4026	0.3652
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.975633	0.5875	0.5330	0.0188	0.0120
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	0.086765	1.0771	1.0512	0.1615	0.0965
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.205796	0.3782	0.3637	0.0499	0.0403
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.051679	1.0005	0.9640	0.1852	0.1276
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.041208	1.9043	1.7533	0.3573	0.2675
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.025236	4.0925	3.7107	0.7666	0.7460
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.032934	3.9588	3.7353	0.4901	0.4055
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.019944	12.3036	10.6818	0.9165	0.9172
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.027351	3.7630	3.5243	0.5881	0.5012
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.017979	9.6504	8.5331	0.8600	0.8467
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	0.087331	1.1936	1.1725	0.1601	0.0969
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.079243	1.1839	1.1693	0.1529	0.1001

The descriptive semi-log fits are

$$n\text{-MAE}_{\mathrm{SchNet}} = 0.203 + 0.286 \ln D_{\mathrm{IQR}}, \qquad R^2 = 0.946, \tag{S5}$$

$$n\text{-MAE}_{\mathrm{GemNet}} = 0.178 + 0.290 \ln D_{\mathrm{IQR}}, \qquad R^2 = 0.905. \tag{S6}$$

Kendall τ is 0.872 for SchNet and 0.821 for GemNet over the 13 target means. These fits summarize the present panel; 39 GNN runs per architecture do not constitute 39 independent target-design observations.

Supplementary Note 7: Normalization and Panel Sensitivity

A high correlation between D_{IQR} and n-MAE can partly arise because both contain an inverse target IQR. We therefore repeat the analysis with unnormalized test MAE, unnormalized test roughness, and test-variance-normalized roughness. All variants use the same test graphs; only normalization changes. DFTB is excluded throughout.

Table S9: Cross-target associations with test error. S: SchNet; G: GemNet. Roughness is logarithm-transformed. DFTB is excluded from every row; N counts targets, not seeds. R^2 is in-sample, not cross-validated.

Comparison	N	S r	S R^2	S τ	G r	G R^2	G τ
Shared test IQR	13	0.972	0.946	0.872	0.951	0.905	0.821
Unnormalized error	13	0.245	0.060	0.231	0.576	0.332	0.487
Unnormalized roughness	13	-0.633	0.401	-0.564	-0.587	0.345	-0.590
Neither axis IQR-normalized	13	0.259	0.067	0.128	-0.137	0.019	-0.154
Variance-normalized roughness	13	0.436	0.190	0.410	0.334	0.111	0.282
DFT-BoE + seven residuals	8	0.966	0.933	0.929	0.954	0.909	0.786
Six descriptor residuals	6	0.919	0.844	0.867	0.905	0.820	0.733

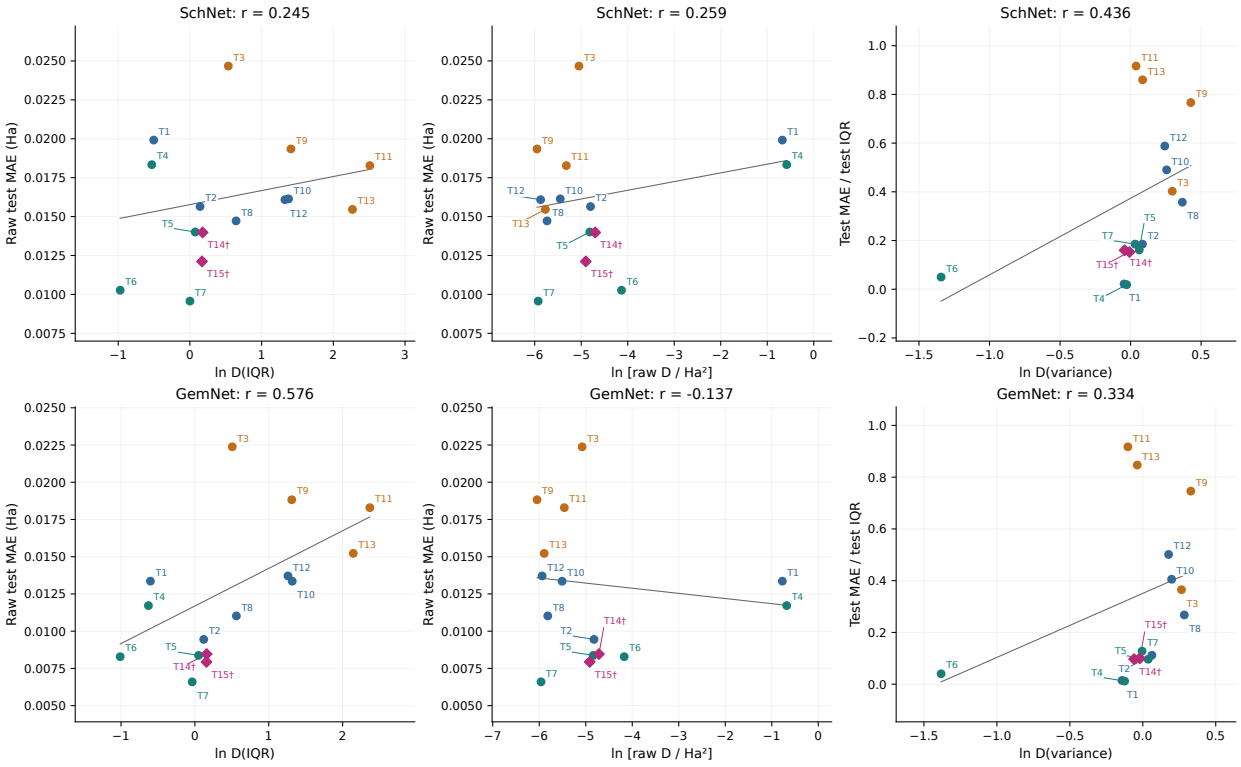


Figure S7: **Controls for shared normalization.** Columns use raw error with IQR-normalized roughness, neither axis IQR-normalized, and variance-normalized roughness with IQR-normalized error. Lines and Pearson r use only the 13 main targets. DFTB† remains a visual control.

The strongest association is not invariant to these choices. With raw error instead of n-MAE, R^2 falls to 0.060 for SchNet and 0.332 for GemNet. With neither axis IQR-normalized, R^2 is 0.067 and 0.019. The normalized fit must not be presented as equally strong evidence that unnormalized roughness predicts absolute error independently of target scale.

Using the same test cohort makes the comparison internally consistent, but does not eliminate shared-denominator coupling. As another descriptive check, we linearly residualize $\ln D_{\mathrm{IQR}}$ and $\ln n$-MAE against $\ln \mathrm{IQR}_{\mathrm{test}}$, including an intercept, and correlate the residual vectors. Partial Pearson correlations are 0.640 and 0.690. This checks association beyond a linear log-scale trend, but does not establish causality or a universal error law in a small, related target panel.

The association table also reports narrower panels: the DFT-BoE parent and its seven non-DFTB residuals, and the six descriptor residuals alone. Leaving out each main target in turn gives semi-log R^2 ranges of 0.927–0.965 for SchNet and 0.872–0.927 for GemNet. These are influence diagnostics, not cross-validated prediction scores.

Molecular neighbourhood information beyond shared normalization

A weaker raw correlation alone does not establish that the normalized association is artificial. We repeat the original control with 5,000 joint test-label permutations (seed 20260914), applying the same molecule permutation to all 13 targets and both architecture proxies. This preserves target distributions, IQRs, and cross-target relationships while disrupting label alignment with molecular neighbourhoods. Graphs, weights, and mean GNN test errors remain fixed; no GNN is retrained. DFTB is excluded. These controls are retrospective, not preregistered.

Table S10: **Corrected-test shared-normalization permutation control.** Fits use 13 primary targets per architecture. Null intervals are the 2.5th–97.5th percentiles over 5,000 joint molecule-label permutations, not confidence intervals for prediction. The one-sided p uses signed Pearson r, not R^2.

Model	Observed R^2	Null median R^2	Null 95% interval	p
GemNet	0.905	0.862	0.834–0.885	0.00020
SchNet	0.946	0.855	0.831–0.879	0.00020

For both models, the observed signed Pearson correlation of $\ln D_{\mathrm{IQR}}$ with test n-MAE exceeds all 5,000 shuffled values: one-sided plus-one $p = 1/5001 \approx 0.00020$, conditional on this panel and randomization scheme, without adjustment for the two architecture comparisons. High shuffled median R^2 values (GemNet: 0.862; SchNet: 0.855) show that target distributions and normalization contribute substantially. The stronger observed associations support additional information from label–geometry alignment beyond this conditional null, not a purely physical explanation or universal learnability law. The observed-minus-shuffled R^2 is not a decomposition of explained variance.

Table S11: **All similar-test-IQR primary target pairs.** The larger IQR is at most 1.10 times the smaller; selection does not use roughness or errors. Values follow target order. Roughness uses PRE for SchNet and PRE+A for GemNet; normalized error is mean test MAE/test IQR. These descriptive comparisons are not independent confirmation on new data.

Model	Targets	IQR (Ha)	D_{IQR}	n-MAE
GemNet	$E_{\mathrm{ref}}^{\mathrm{DFT}} \to E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.91832 → 0.97563	0.548 → 0.533	0.015 → 0.012
GemNet	$E_{\mathrm{ref,BoE}}^{\mathrm{DFT}} \to E_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.08450 → 0.08677	1.127 → 1.051	0.112 → 0.096
GemNet	$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}} \to \Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.02524 → 0.02735	3.711 → 3.524	0.746 → 0.501
SchNet	$E_{\mathrm{ref}}^{\mathrm{DFT}} \to E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.91832 → 0.97563	0.604 → 0.587	0.022 → 0.019
SchNet	$E_{\mathrm{ref,BoE}}^{\mathrm{DFT}} \to E_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.08450 → 0.08677	1.152 → 1.077	0.185 → 0.161
SchNet	$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}} \to \Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.02524 → 0.02735	4.092 → 3.763	0.767 → 0.588

The original 10% IQR-matching threshold selects three pairs, all concordant in roughness and normalized-error ordering for both models. Linear PRE+A has 8.4% larger IQR than MLP-BoB, but lower roughness and normalized error. Matching narrows, but does not eliminate, denominator differences. Together, these controls support a geometry-sensitive component while retaining the shared-normalization limitation; they do not establish causality.

Supplementary Section 5: DFTB Semiempirical Control

Supplementary Note 8: DFTB Coverage and Matched Evaluation

DFTB tests whether another semiempirical baseline can yield a useful residual. It remains a supporting control because successful calculations cover a different cohort and only seed 101 was trained. Of 71,038 neutral tmQM molecules, 362 lack successful DFTB calculations: 290 training, 36 validation, and 36 test molecules. Valid splits contain 56,540, 7,068, and 7,068 molecules. Ten QM9 DFTB failures leave 130,797 valid molecules. Failed energies are not imputed.

Raw DFTB calculations and fixed standard-molecule reference inputs were reused; the learned composition correction was newly fitted on successful tmQM training molecules only. Historical fitted residuals were not used. The DFT parent is the canonical train-only DFT-BoE target restricted to the valid mask. Frozen DFTB coefficients generate validation, test, and QM9 targets.

On common DFTB-valid test molecules, the residual's D_{IQR} is 1.184 in PRE and 1.169 in PRE+A, compared with 1.127 and 1.113 for the DFT parent under identical masks. This is a paired target-roughness comparison, not a comparison across different graph cohorts. The DFTB residual is slightly rougher than the parent in both proxies, despite improving matched-subset test MAE; lower normalized roughness is therefore not a necessary condition for lower absolute error. DFTB energy and residual GNNs reach test MAEs of 0.013980 and 0.012119 Ha for SchNet, and 0.008462 and 0.007936 Ha for GemNet. Direct DFTB energy and DFT residual remain different prediction tasks.

Table S12: Seed-101 DFT reconstruction on exactly matched DFTB-valid evaluation molecules (Ha). Shifted errors are oracle diagnostics. Training cohorts remain different: DFTB used only successful training molecules. These rows are not pooled into any primary analysis.

Dataset	Route	N	S raw	S shifted	G raw	G shifted
tmqm	$E^{\mathrm{DFT}}_{\mathrm{ref,BoE\text{-}lin}}$	7,068	0.015563	0.015623	0.009794	0.009832
tmqm	$\Delta^{\mathrm{xTB}}_{\mathrm{ref,BoE}}$	7,068	0.009499	0.009480	0.006538	0.006560
tmqm	$\Delta^{\mathrm{DFTB\dagger}}_{\mathrm{ref,BoE}}$	7,068	0.012119	0.012125	0.007936	0.007954
qm9	$E^{\mathrm{DFT}}_{\mathrm{ref,BoE\text{-}lin}}$	130,797	0.252018	0.016036	0.255477	0.019912
qm9	$\Delta^{\mathrm{xTB}}_{\mathrm{ref,BoE}}$	130,797	0.024872	0.006813	0.024622	0.006151
qm9	$\Delta^{\mathrm{DFTB\dagger}}_{\mathrm{ref,BoE}}$	130,797	0.053349	0.011783	0.056495	0.012578

The matched table evaluates seed-101 direct-DFT, xTB-residual, and DFTB-residual predictors on identical DFTB-valid test or QM9 molecules. This removes evaluation-mask differences, but does not equalize their training cohorts. No separately trained exact-DFTB-mask direct-DFT GNN is included in the completed 82-run matrix. The control therefore supports a qualified comparison, not an unrestricted DFTB ranking or a three-seed uncertainty estimate.

Supplementary Section 6: Independent-Seed Results

Supplementary Note 9: GNN Seed Variation

The following tables report every main-run test and raw QM9 MAE, not only a representative seed. Each seed has its own validation-selected checkpoint. For $S = 3$, a metric m_s is summarized by

$$\bar{m} = \frac{1}{S}\sum_s m_s, \qquad s_m = \sqrt{\frac{1}{S-1}\sum_s (m_s - \bar{m})^2}. \tag{S7}$$

This SD measures training-seed variation for a fixed dataset and target. It is not uncertainty from drawing new molecules, changing the baseline fitting seed, or selecting a different domain. Deterministic IQR and roughness values have no GNN-seed error bars.

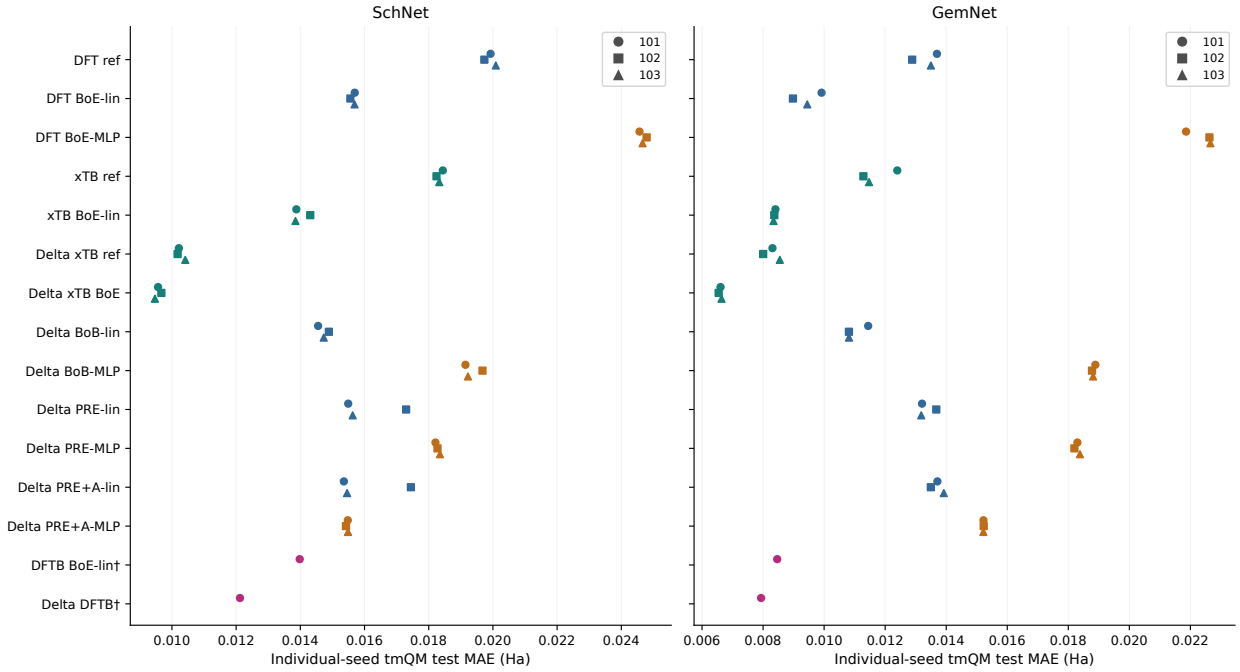


Figure S8: **Individual-seed tmQM test errors.** Circles, squares, and triangles show seeds 101, 102, and 103. All main targets have three runs per architecture. DFTB†, in magenta, has seed 101 only.

Table S13: SchNet individual-seed tmqm native-target MAEs (Ha). All predictions use each run's best-validation checkpoint. A dash denotes an intentionally unrun DFTB seed, not a zero error.

Target	101	102	103
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	0.019928	0.019733	0.020089
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.015694	0.015559	0.015690
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.024566	0.024789	0.024662
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.018439	0.018242	0.018328
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	0.013876	0.014309	0.013849
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.010216	0.010176	0.010416
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.009568	0.009671	0.009469
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.014554	0.014888	0.014729
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.019145	0.019675	0.019222
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.015492	0.017296	0.015631
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.018210	0.018273	0.018350
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.015357	0.017444	0.015456
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.015479	0.015422	0.015485
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	0.013980	–	–
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.012119	–	–

Table S14: GemNet individual-seed tmqm native-target MAEs (Ha). All predictions use each run's best-validation checkpoint. A dash denotes an intentionally unrun DFTB seed, not a zero error.

Target	101	102	103
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	0.013696	0.012885	0.013498
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.009917	0.008979	0.009448
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.021859	0.022624	0.022656
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.012397	0.011287	0.011471
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	0.008404	0.008368	0.008343
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.008307	0.008003	0.008548
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.006606	0.006544	0.006638
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.011443	0.010813	0.010816
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.018887	0.018777	0.018813
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.013208	0.013675	0.013182
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.018298	0.018200	0.018381
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.013711	0.013497	0.013921
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.015221	0.015229	0.015217
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	0.008462	–	–
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.007936	–	–

Table S15: SchNet individual-seed qm9 native-target MAEs (Ha). All predictions use each run's best-validation checkpoint. A dash denotes an intentionally unrun DFTB seed, not a zero error.

Target	101	102	103
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	1.006303	1.005030	1.015885
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.252019	0.258079	0.259814
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.483568	0.465546	0.471873
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.942838	0.944062	0.941814
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	0.274410	0.277747	0.271944
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.055792	0.055150	0.058758
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.024874	0.025531	0.021646
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.273621	0.279276	0.276816
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.261957	0.266765	0.263364
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.272982	0.277887	0.275026
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.213049	0.214906	0.214518
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.258401	0.256111	0.255926
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.215360	0.215299	0.215430
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	0.306405	–	–
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.053349	–	–

Table S16: GemNet individual-seed qm9 native-target MAEs (Ha). All predictions use each run's best-validation checkpoint. A dash denotes an intentionally unrun DFTB seed, not a zero error.

Target	101	102	103
$E_{\mathrm{ref}}^{\mathrm{DFT}}$	1.055482	1.044236	1.080928
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFT}}$	0.255478	0.244060	0.245231
$E_{\mathrm{ref,BoE\text{-}MLP}}^{\mathrm{DFT}}$	0.476675	0.475178	0.483564
$E_{\mathrm{ref}}^{\mathrm{xTB}}$	0.986678	0.992279	0.986614
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{xTB}}$	0.283163	0.281808	0.276816
$\Delta_{\mathrm{ref}}^{\mathrm{xTB}}$	0.055135	0.054089	0.049559
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{xTB}}$	0.024624	0.025071	0.024750
$\Delta_{\mathrm{lin}}^{\mathrm{BoB}}$	0.262610	0.266210	0.267735
$\Delta_{\mathrm{MLP}}^{\mathrm{BoB}}$	0.257258	0.254821	0.246993
$\Delta_{\mathrm{lin}}^{\mathrm{PRE}}$	0.262614	0.263540	0.264543
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE}}$	0.213722	0.214740	0.215478
$\Delta_{\mathrm{lin}}^{\mathrm{PRE+A}}$	0.253586	0.252567	0.257962
$\Delta_{\mathrm{MLP}}^{\mathrm{PRE+A}}$	0.214595	0.214208	0.215065
$E_{\mathrm{ref,BoE\text{-}lin}}^{\mathrm{DFTB\dagger}}$	0.306466	–	–
$\Delta_{\mathrm{ref,BoE}}^{\mathrm{DFTB\dagger}}$	0.056495	–	–

SchNet linear PRE and PRE+A show larger test variability than most other targets. Their seed-102 test MAEs are 0.017296 and 0.017444 Ha, compared with approximately 0.0154–0.0156 Ha for the other seeds. These runs are retained rather than discarded. Differences between mean scores should not be described as decisive when comparable to this variation.

Baseline Fitting Seeds Are Separate

A baseline-only check repeated each MLP fit with seeds 102 and 103, holding architecture, inputs, train-only preprocessing, and checkpoint selection fixed. Baseline seed 101 remains canonical; predictions and residual targets were not averaged over fitting seeds.

Table S17: Separate MLP-baseline fitting-seed check (Ha), seeds 101, 102, and 103. These are not the GNN seeds. Only baseline seed 101 defines the targets used by the 78 main GNN runs.

Descriptor	Test MAE	Range
BOB	0.021219 ± 0.000092	0.021124–0.021307
BOE	0.045442 ± 0.000512	0.044907–0.045926
PRE	0.018639 ± 0.000120	0.018509–0.018746
PRE+A	0.016523 ± 0.000201	0.016335–0.016735

Baseline test MAE SD is below 0.0006 Ha in each family, but individual residual labels are not identical. Paired test residuals show rare large differences relative to the canonical fit, with maxima of 1.093 Ha for PRE and 0.866 Ha for PRE+A. These scores use the already stored, frozen test predictions without refitting or selecting among baseline seeds. The present GNN error bars do not quantify this additional variability. Robustness to baseline fitting seed would require downstream evaluation on those alternative targets.

Supplementary Section 7: Supplementary Methods

Datasets, References, and Splits

The neutral tmQM benchmark has 71,038 molecules. Geometries derive from tmQM, and DFT energies are at ωB97M-V/def2-SVPD [1, 2, 3]. QM9 DFT energies use the same level, evaluated with Psi4 [4]. QM9 supplies no fitted baseline or GNN parameters.

Frozen split identifiers were retained during target regeneration. Joins use molecule IDs or explicitly mapped SIDs; CSV, LMDB, and prediction-array orders are not assumed to coincide. The corrected registry records exact split-file checksums. A filename containing *full* denotes concatenated split rows evaluated with one frozen model, not a model fitted on all molecules.

The fixed method-specific atom-reference convention is

$$E^m_{\mathrm{ref}}(i) = E^m_{\mathrm{raw}}(i) - \sum_Z n_Z(i)\epsilon^m_Z. \tag{S8}$$

Here ϵ^m_Z belongs to an external reference convention, not a regression fitted to held-out molecular energies. Canonical DFT E_{ref} values were retained as authoritative input; this is distinct from reusing a historical molecular-label-fitted composition reference.

Train-Only Baseline Fitting

Linear BoE predicts atom-referenced energy from 44 elemental counts, with an intercept. Ridge regularization is selected by five-fold shuffled training-only cross-validation, split seed 42, and mean fold-held-out MAE. Standardization is refitted within each fold. Final coefficients use training data only and are frozen for validation, test, and transfer. DFT, xTB, and DFTB have separate BoE fits; DFTB uses only successful training molecules.

The six descriptor baselines predict $Y = E^{\mathrm{DFT}}_{\mathrm{ref,BoE\text{-}lin}}$, giving $\Delta^b = Y - \hat{E}_b$. BoB is a 4,950-dimensional RDKit typed-bond-count descriptor, not conventional Coulomb-matrix Bag-of-Bonds. PRE has 34,650 pair-radial features, 35 Gaussian channels, and a 6.0 Å cutoff. Centres span 0.8–3.0 Å at 0.1 Å intervals (width 0.12 Å), then 3.25–6.0 Å at 0.25 Å intervals (width 0.30 Å). PRE+A adds 11,088 angular features, giving 45,738 total: 44 central elements, 21 unordered pairs of six radial shells, and 12 $\cos\theta$ bins.

Linear descriptor models remove training-zero-variance channels and divide by training population SD without centring. Filtering and scaling are refitted in every cross-validation fold. The descriptor ridge grid has 25 logarithmic values from 10^{-6} to 10^{6}; selected values must not be grid endpoints. Final fits use only the descriptor-common training split.

The four MLP baselines use hidden widths 512 and 128, SiLU, dropout 0.10, and a scalar linear output. MLP-BoE predicts $E^{\mathrm{DFT}}_{\mathrm{ref}}$ from counts; descriptor MLPs predict Y. AdamW uses learning rate 10^{-3}, weight decay 10^{-4}, SmoothL1 loss with $\beta = 0.10$, batch size 1024, clipping at 1, at most 800 epochs, and early-stopping patience 80 with minimum improvement 10^{-5} Ha. Training data fit parameters and preprocessing; validation selects the lowest-MAE checkpoint. Thus training-only fitting does not mean validation labels were never used for model selection.

Semiempirical Targets

GFN2-xTB supplies the xTB baseline [5]. Both referencing levels are included:

$$\Delta^{\mathrm{xTB}}_{\mathrm{ref}} = E^{\mathrm{DFT}}_{\mathrm{ref}} - E^{\mathrm{xTB}}_{\mathrm{ref}}, \tag{S9}$$

$$\Delta^{\mathrm{xTB}}_{\mathrm{ref,BoE}} = E^{\mathrm{DFT}}_{\mathrm{ref,BoE}} - E^{\mathrm{xTB}}_{\mathrm{ref,BoE}}. \tag{S10}$$

DFTB is analogous after success filtering. DFTB+ uses SCC-DFTB/DFTB2 with PTBP Slater–Koster parameters, without third-order or dispersion corrections [6, 7, 8]. This broad parameterization is a control, not a claim of optimal parameters for the molecular domain.

GNN Configurations and Selection

Both architectures use energy-only learning, a 6.0 Å cutoff, on-the-fly graphs, no periodic boundary conditions, and no force regression. SchNet uses 128 hidden channels, 128 filters, six interactions, 50 Gaussian functions, and sum readout [9]. GemNet-T uses three blocks, atom and edge widths 512, triplet and bilinear-triplet widths 64, radial and angular embedding widths 16, 128 radial functions, seven spherical functions, at most 50 neighbours, SiLU, and extensive summation [10]. It uses Gaussian radial functions, spherical harmonics, and a polynomial envelope with exponent five.

AdamW with AMSGrad starts at 5×10^{-4}. ReduceLROnPlateau uses factor 0.8 and patience three. Configurations specify MSE energy loss, EMA decay 0.999, clipping at 10, validation every 5,000 steps, and an 80-epoch budget. SchNet batch size is 16; GemNet batch size is four, including evaluation. Validation controls scheduling and checkpoint selection; test and QM9 labels do not.

Each target is standardized by its own training mean and population SD, shared between architectures and seeds. These were independently recalculated and checked against all 82 configurations, and are never fitted on validation, test, or QM9.

Table S18: Target scales in Ha. Training means and population standard deviations were independently checked against all 82 configuration files and remain unchanged. Test IQR normalizes reported test errors and test-graph roughness; it is not used for training.

Target	μ_{train}	σ_{train}	IQR_{test}
$E_{\text{ref}}^{\text{DFT}}$	1.035182	0.733244	0.918321
$E_{\text{ref,BoE-lin}}^{\text{DFT}}$	0.000000	0.091165	0.084504
$E_{\text{ref,BoE-MLP}}^{\text{DFT}}$	0.005657	0.065718	0.061288
$E_{\text{ref}}^{\text{xTB}}$	0.986222	0.764581	0.975633
$E_{\text{ref,BoE-lin}}^{\text{xTB}}$	0.000000	0.095927	0.086765
$\Delta_{\text{ref}}^{\text{xTB}}$	0.048960	0.251295	0.205796
$\Delta_{\text{ref,BoE}}^{\text{xTB}}$	0.000000	0.055191	0.051679
$\Delta_{\text{lin}}^{\text{BoB}}$	0.000000	0.051867	0.041208
$\Delta_{\text{MLP}}^{\text{BoB}}$	0.001850	0.029471	0.025236
$\Delta_{\text{lin}}^{\text{PRE}}$	-0.000000	0.040629	0.032934
$\Delta_{\text{MLP}}^{\text{PRE}}$	0.001371	0.011373	0.019944
$\Delta_{\text{lin}}^{\text{PRE+A}}$	-0.000000	0.033210	0.027351
$\Delta_{\text{MLP}}^{\text{PRE+A}}$	0.002017	0.010266	0.017979
$E_{\text{ref,BoE-lin}}^{\text{DFTB†}}$	0.000000	0.097167	0.087331
$\Delta_{\text{ref,BoE}}^{\text{DFTB†}}$	-0.000748	0.085898	0.079243

Roughness and Statistical Scope

Standardized roughness graphs remove zero-variance channels on the test graph cohort and divide retained channels by that cohort's population SD without centring. This unsupervised transformation is distinct from train-only regressor preprocessing. Cosine distances between normalized vectors define neighbours; unique undirected union edges receive

$$w_{ij} = \exp[-d_{ij}^2/(\sigma_i \sigma_j)], \tag{S11}$$

where σ_i is the tenth-neighbour distance. The common test graph has 51,535 PRE or 53,533 PRE+A edges. Every main target uses identical edges and weights within an architecture. DFTB and its paired DFT parent use separately constructed valid-subset test graphs, with 51,268 PRE or 53,246 PRE+A edges.

IQRs use linearly interpolated quartiles; variance normalization uses population variance. The primary fit is unweighted ordinary least squares across 13 target means, with an intercept and $\ln D_{\mathrm{IQR}}$ predictor. Pearson r, in-sample R^2, Spearman ρ, and Kendall τ are descriptive. Targets are related transformations of the same molecules, so target-wise independence is not assumed. An explicit 13-target whitelist excludes DFTB before every pooled calculation.

This corrected analysis reports two proxy graphs, normalization controls, panel-sensitivity checks, and 5,000 new joint test-label permutations with similar-IQR comparisons. Historical graph-grid results and synthetic-network rescaling fits are not presented as corrected-data experiments. The new permutation values use the corrected test protocol, not the historical validation results. The association does not establish a causal mechanism or guaranteed prospective ranking.

Prediction Audit and Reproducibility

The inventory contains 82 unique runs with complete predictions for both evaluation datasets. The SI audit re-read prediction arrays, verified unique IDs and complete available-target coverage, and recomputed all 164 native-target MAEs. The largest difference from the final ledgers was 6.7×10^{-16} Ha. All split target files matched corrected-registry checksums, split molecule sets were disjoint, and test and QM9 checkpoint paths matched. The final tmQM ledger records checkpoint checksums; the SI audit also records prediction and input checksums.

The SI bundle contains the target registry, final ledgers, target statistics, complete per-seed metrics, baseline scores, test-graph edges and weights, roughness values, correlation inclusion masks, checksums, and plotting code. Raw and oracle-shifted RMSE, R^2, Pearson correlation, Spearman correlation, Kendall correlation, and bias remain available in machine-readable tables where the printed tables select fewer metrics. These materials are distributed with the corrected code and data release, `v0.2.0-corrected` [11]. Raw structures, LMDB databases, model checkpoints, and full prediction arrays are not included in that release. Rebuilding this SI neither retrains models nor changes the main manuscript.

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